Historia Módulos de Memoria

- DIP: Dual In-line Package



- SIPP: Single In-line Pin Package



- SIMM (Single In-line Memory Module): de 30 contactos, **8 bits**



- SIMM de72 contactos, **32 bits**



- DIMM, Dual In-line Memory Module **64-bit**, 168 contactos, SDR SDRAM



- DIMM 184 contactos, DDR SDRAM



- DIMM 240 contactos, DDR2 SDRAM

Módulos de Memoria

DDR



DDR2



DDR3

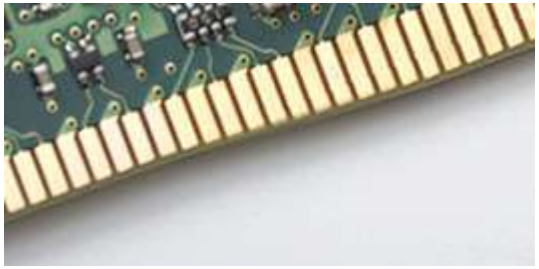


DDR4



DDR5





DDR4 y DDR5 ligeramente curvada: mejor inserción y menos estrés en PBC

https://www.kingston.com/en/memory/ddr4-overview

Denominación Módulos de Memoria

- Even though the term DDR*x-yyy* is really just for individual DDR chips and the term PC*x-zzzz* is for DDR sticks, this tradition of two names for every speed of RAM is a bit of a challenge because you'll often hear both terms used interchangeably
  - DDR*x-yyy*: *x* versión, *yyy* MTransferencias/s
  - PC*x-zzzz*: : *x* versión, *zzzz* MBytes/s
  - Una transferencia DIMM son 64 bits de datos.

Página 184: http://www.houstonisd.org/cms/lib2/tx01001591/centricity/domain/10450/chap6.pdf
http://www.hardwaresecrets.com/everything-you-need-to-know-about-ddr-ddr2-and-ddr3-memories/

Memoria SDRAM

Memory configurations

You may install 2 GB, 4 GB, 8 GB, 16 GB, and 32 GB unbuffered and non-ECC DDR4 DIMMs into the DIMM sockets.



You may install varying memory sizes in Channel A and Channel B. The system maps the total size of the lower-sized channel for the dual-channel configuration. Any excess memory from the higher-sized channel is then mapped for single-channel operation.



- The default memory operation frequency is dependent on its Serial Presence Detect (SPD), which is the standard way of accessing information from a memory module. Under the default state, some memory modules for overclocking may operate at a lower frequency than the vendor-marked value.
- For system stability, use a more efficient memory cooling system to support a full memory load or overclocking condition.
- Always install the DIMMS with the same CAS Latency. For an optimum compatibility, we recommend that you install memory modules of the same version or data code (D/C) from the same vendor. Check with the vendor to get the correct memory modules.
- Visit the ASUS website for the latest QVL.

Memoria SDRAM

DDR Speeds

Clock Speed	DDR Speed Rating	PC Speed Rating
100 MHz	DDR200	PC1600
133 MHz	DDR266	PC2100
166 MHz	DDR333	PC2700
200 MHz	DDR400	PC3200
217 MHz	DDR433	PC3500
233 MHz	DDR466	PC3700
250 MHz	DDR500	PC4000
275 MHz	DDR550	PC4400
300 MHz	DDR600	PC4800

En cada tipo de memoria hay 2 frecuencias y 1 tasa de transferencia

DDR2 Speeds

Core RAM Clock Speed	DDR I/O Speed	DDR2 Speed Rating	PC Speed Rating
100 MHz	200 MHz	DDR2-400	PC2-3200
133 MHz	266 MHz	DDR2-533	PC2-4200
166 MHz	333 MHz	DDR2-667	PC2-5300
200 MHz	400 MHz	DDR2-800	PC2-6400
250 MHz	500 MHz	DDR2-1000	PC2-8000

DDR3 Speeds

Core RAM Clock Speed	DDR I/O Speed	DDR3 Speed Rating	PC Speed Rating
100 MHz	400 MHz	DDR3-800	PC3-6400
133 MHz	533 MHz	DDR3-1066	PC3-8500
166 MHz	667 MHz	DDR3-1333	PC3-10667
200 MHz	800 MHz	DDR3-1600	PC3-12800

Página 184: http://www.houstonisd.org/cms/lib2/tx01001591/centricity/domain/10450/chap6.pdf
http://www.hardwaresecrets.com/everything-you-need-to-know-about-ddr-ddr2-and-ddr3-memories/

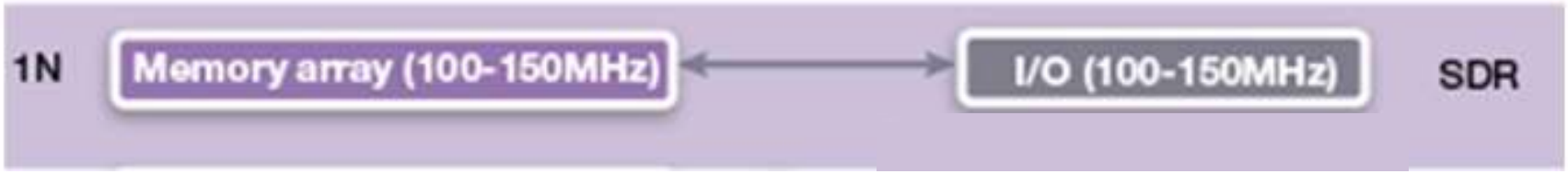
Memoria SDRAM

- Clock Speed, Core RAM Clock Speed: frecuencia del reloj interno de la matriz SDRAM, celdas y circuitos internos de la SDRAM trabajan tomando como referencia ese reloj.
- DDRx I/O Speed: Es la frecuencia de la interfaz de entrada/salida entre el chip de memoria y el controlador de memoria. tCL, tRCD, tRP y tRAS se refieren a esta frecuencia.
- DDRx I/O Rating: indica tasa de transferencias, como es DDR, en cada ciclo 2 transferencias.

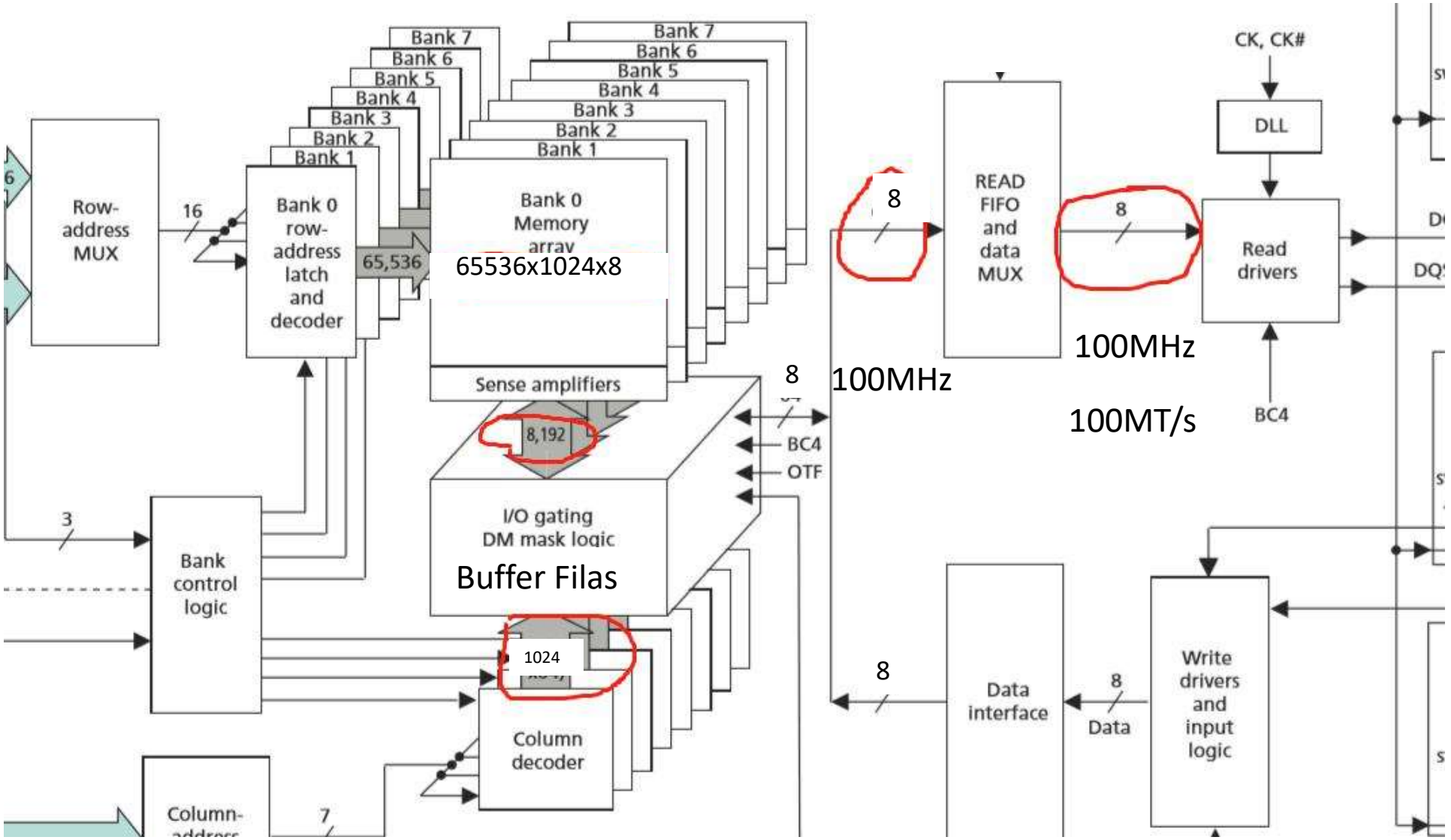
Memoria SDRAM

- SDR SDRAM: Single Data Rate SDRAM
- Transfiere 1 dato por ciclo de reloj

Nombre estándar	Frecuencia del bus	Frecuencia de la memoria	Datos transferidos por segundo	Nombre del módulo	Máxima capacidad de transferencia
SDR-66	66,6 MHz	66,6 MHz	66 Millones	PC-66	533 MB/s
SDR-100	100 MHz	100 MHz	100 Millones	PC-100	800 MB/s
SDR-133	133,3 MHz	133,3 MHz	133 Millones	PC-133	1066 MB/s



Chip SDR


Bank 7
Bank 6
Bank 5
Bank 4
Bank 3
Bank 2
Bank 1
Row-address MUX
16
Bank 0 row-address latch and decoder
65,536
Bank 0 Memory array
65536x1024x8
Sense amplifiers
8,192
I/O gating
DM mask logic
Buffer Filas
1024
Bank control logic
3
Column decoder
Column-
7
8
BC4
OTF
READ FIFO and data MUX
8
100MHz
8
100MHz
100MT/s
CK, CK#
DLL
Read drivers
BC4
DQ
Write drivers and input logic
8
Data
Data interface
8

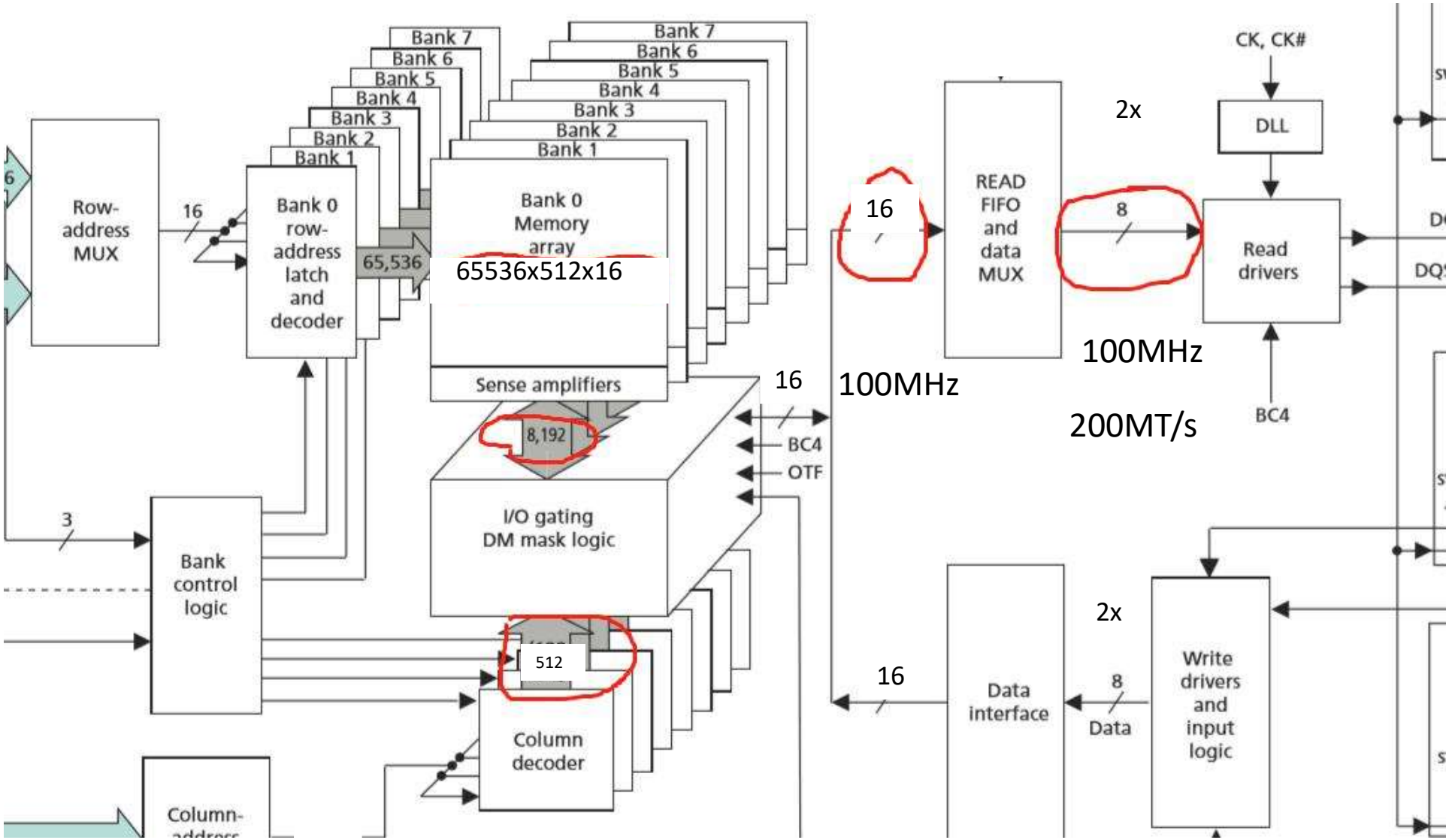
Memoria SDRAM

- DDR SDRAM: Doble Data Rate SDRAM
- Transfiere 2 datos por ciclo de reloj

Standard name	Memory clock (MHz)	Cycle time (ns)	I/O bus clock (MHz)	Data rate (MT/s)	Module name	Peak transfer rate (MB/s)
DDR-200	100	10	100	200	PC-1600	1600
DDR-266	133⅓	7.5	133⅓	266⅔	PC-2100	2133⅓
DDR-333	166⅔	6	166⅔	333⅓	PC-2700	2666⅔
DDR-400	200	5	200	400	PC-3200	3200



Chip DDR

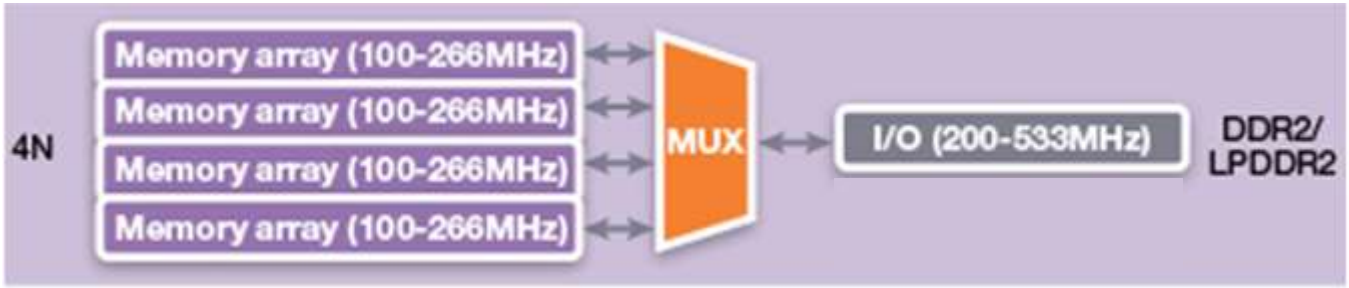

Bank 7
Bank 6
Bank 5
Bank 4
Bank 3
Bank 2
Bank 1
Row-address MUX
16
Bank 0 row-address latch and decoder
65,536
Bank 0 Memory array
65536x512x16
Sense amplifiers
8,192
I/O gating DM mask logic
Bank control logic
3
512
Column decoder
Column-
16
READ FIFO and data MUX
2x
8
CK, CK#
DLL
Read drivers
DQ
100MHz
200MT/s
BC4
OTF
Data interface
Data
Write drivers and input logic

Memoria SDRAM

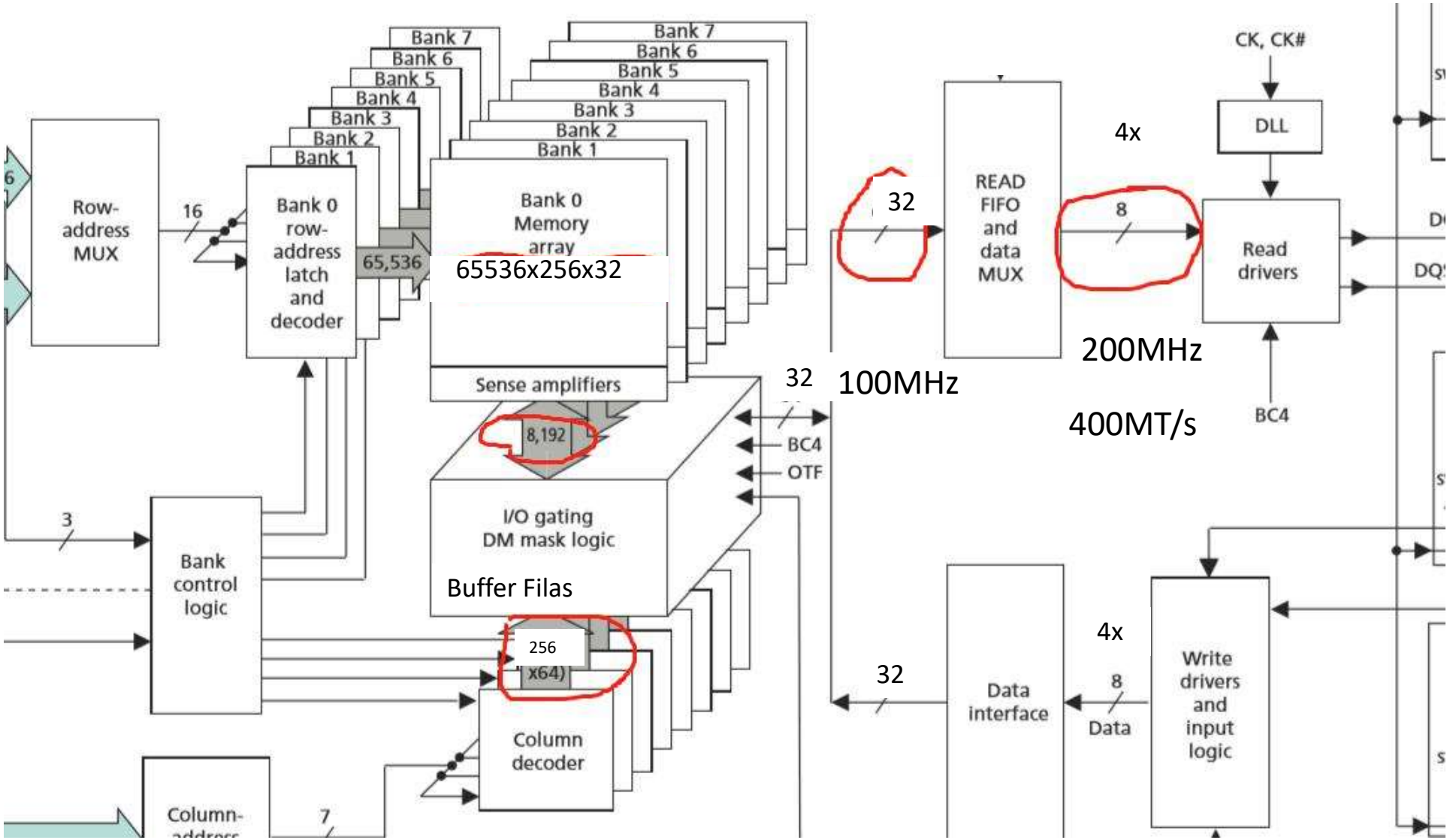
- La DDR2 duplica la frecuencia del reloj del bus
- Transfiere 4 datos por ciclo de reloj

Standard name	Memory clock (MHz)	Cycle time (ns)	I/O bus clock (MHz)	Data rate (MT/s)	Module name	Peak transfer rate (MB/s)
DDR2-400	100	10	200	400	PC2-3200	3200
DDR2-533	133⅓	7½	266⅔	533⅓	PC2-4200*	4266⅔
DDR2-667	166⅔	6	333⅓	666⅔	PC2-5300*	5333⅓
DDR2-800	200	5	400	800	PC2-6400	6400
DDR2-1066	266⅔	3¾	533⅓	1066⅔	PC2-8500*	8533⅓

https://es.wikipedia.org/wiki/DDR2_SDRAM



DDR2


Bank 7
Bank 6
Bank 5
Bank 4
Bank 3
Bank 2
Bank 1
Row-address MUX
16
Bank 0 row-address latch and decoder
65,536
Bank 0 Memory array
65536x256x32
Sense amplifiers
8,192
I/O gating
DM mask logic
Buffer Filas
256
x64)
Bank control logic
3
Column decoder
Column-
7
32
BC4
OTF
READ FIFO and data MUX
4x
8
200MHz
100MHz
400MT/s
CK, CK#
DLL
Read drivers
BC4
DQ
Data interface
Write drivers and input logic
Data

Memoria SDRAM

- La DDR3 duplica la frecuencia de la DDR2
- Transfiere 8 datos por ciclo de reloj

Standard name	Memory clock (MHz)	Cycle time (ns)	I/O bus clock (MHz)	Data rate (MT/s)	Module name	Peak transfer rate (MB/s)
DDR3-800	100	10	400	800	PC3-6400	6400
DDR3-1066	133⅓	7½	533⅓	1066⅔	PC3-8500	8533⅓
DDR3-1333	166⅔	6	666⅔	1333⅓	PC3-10600	10666⅔
DDR3-1600	200	5	800	1600	PC3-12800	12800
DDR3-1866	233⅓	4²⁄₇	933⅓	1866⅔	PC3-14900	14933⅓
DDR3-2133	266⅔	3¾	1066⅔	2133⅓	PC3-17000	17066⅔

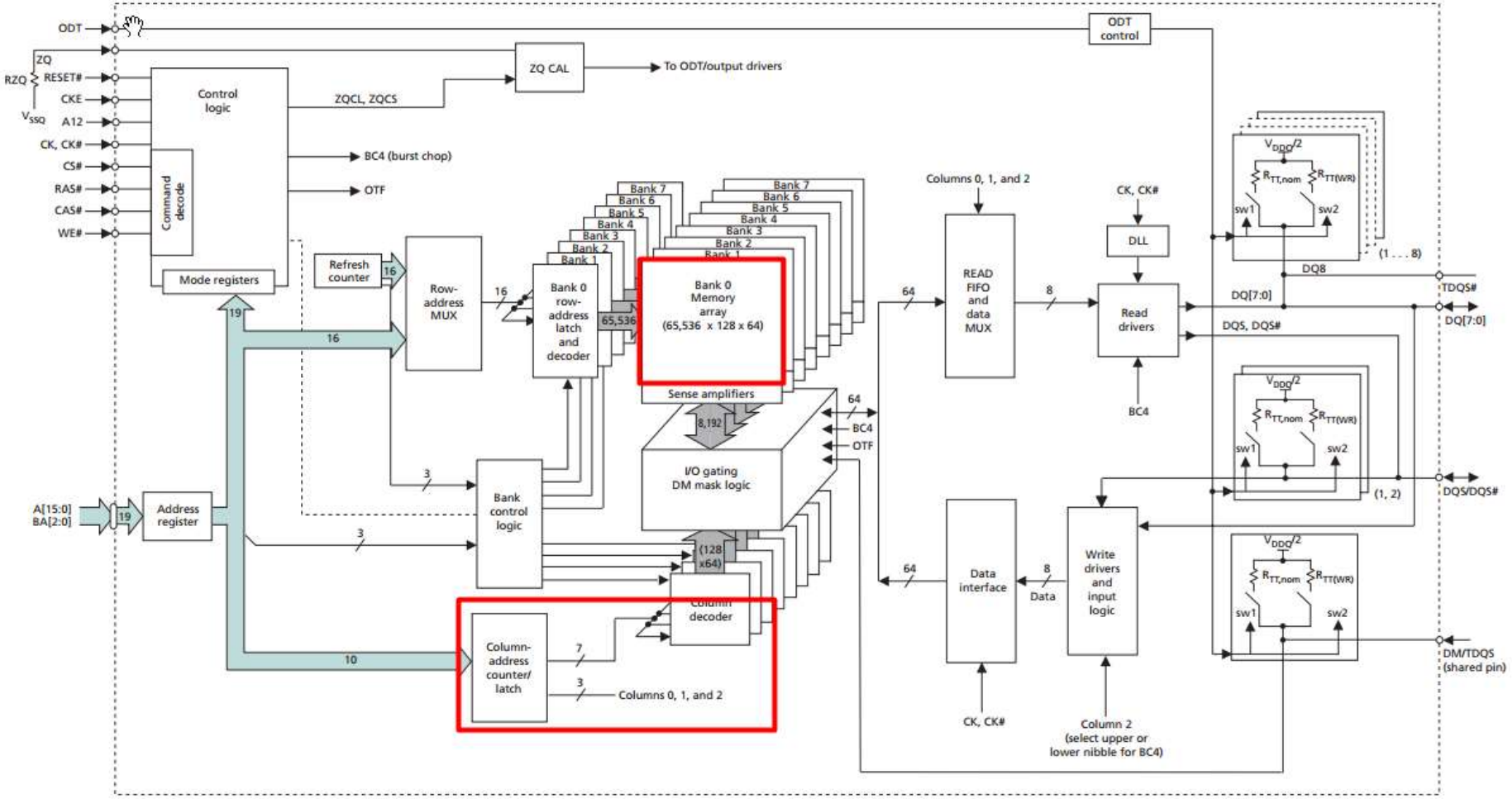


https://es.wikipedia.org/wiki/DDR3_SDRAM

https://www.programmersought.com/article/59963608748/

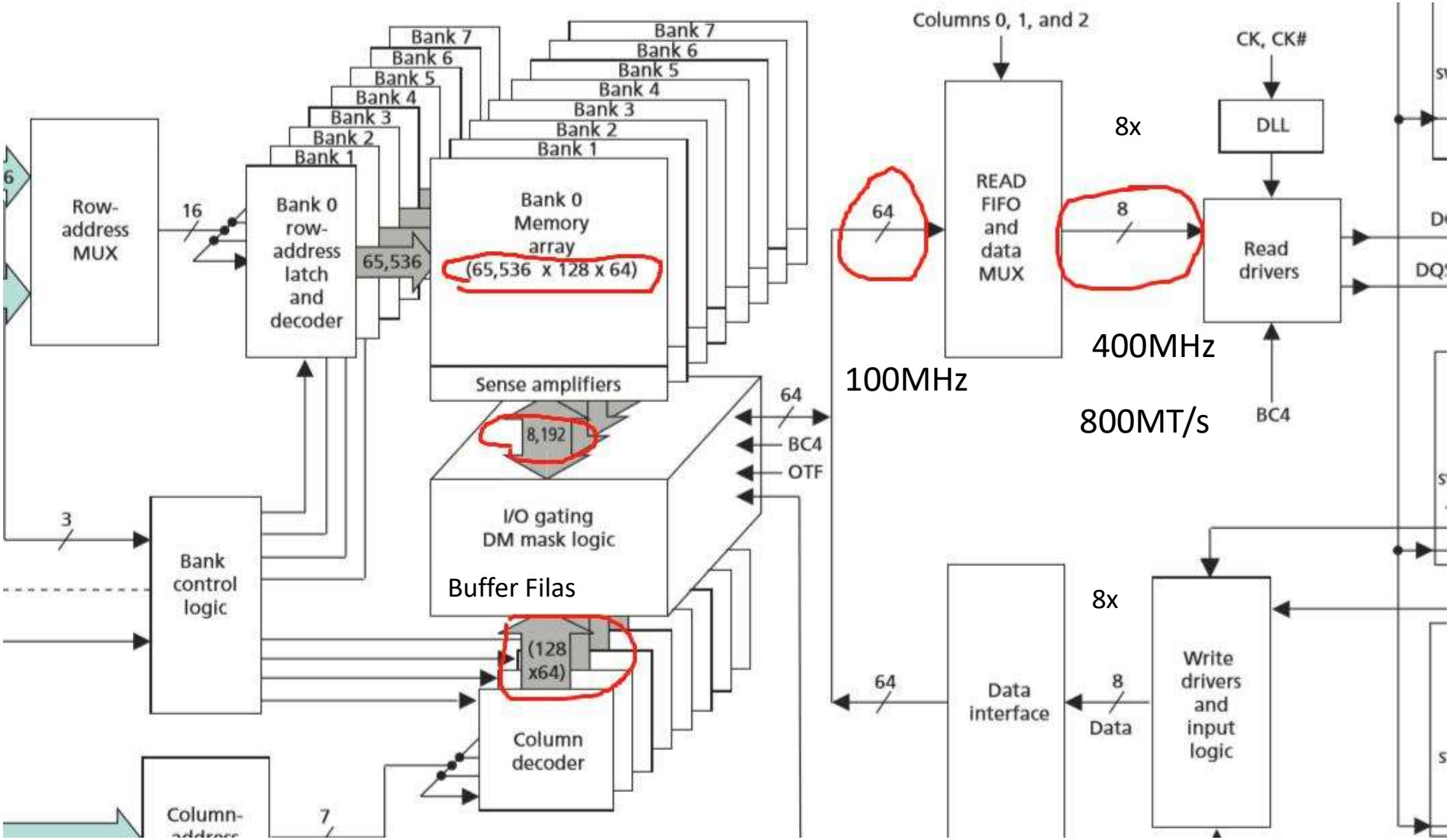
Chip DDR3

Figure 4: 512 Meg x 8 Functional Block Diagram



https://www.programmersought.com/article/59963608748/

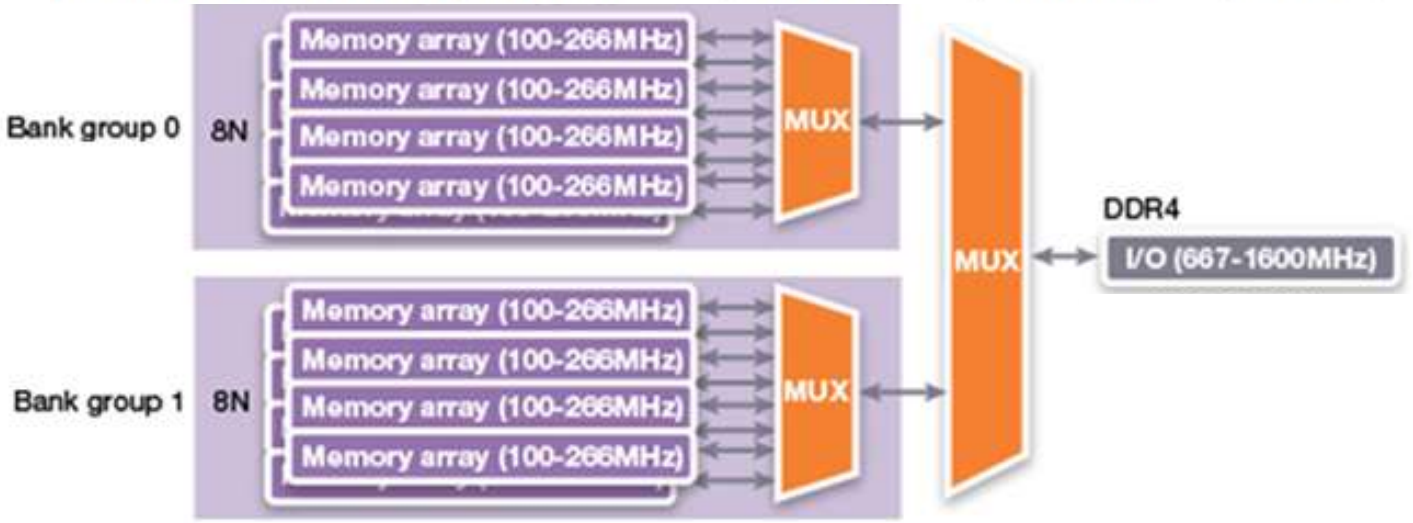
Chip DDR3



Memoria síncrona SDRAM

- DDR4: 16 datos cada T

Chip	Module	Memory Clock	I/O Bus Clock	Transfer rate	Voltage
DDR4-1600	PC4-12800	200 MHz	800 MHz	1600 MT/s	1.2 V
DDR4-1866	PC4-14900	233 MHz	933 MHz	1866 MT/s	1.2 V
DDR4-2133	PC4-17000	266 MHz	1066 MHz	2133 MT/s	1.2 V
DDR4-2400	PC4-19200	300 MHz	1200 MHz	2400 MT/s	1.2 V
DDR4-2666	PC4-21300	333 MHz	1333 MHz	2666 MT/s	1.2 V
DDR4-3200	PC4-25600	400 MHz	1600 MHz	3200 MT/s	1.2 V

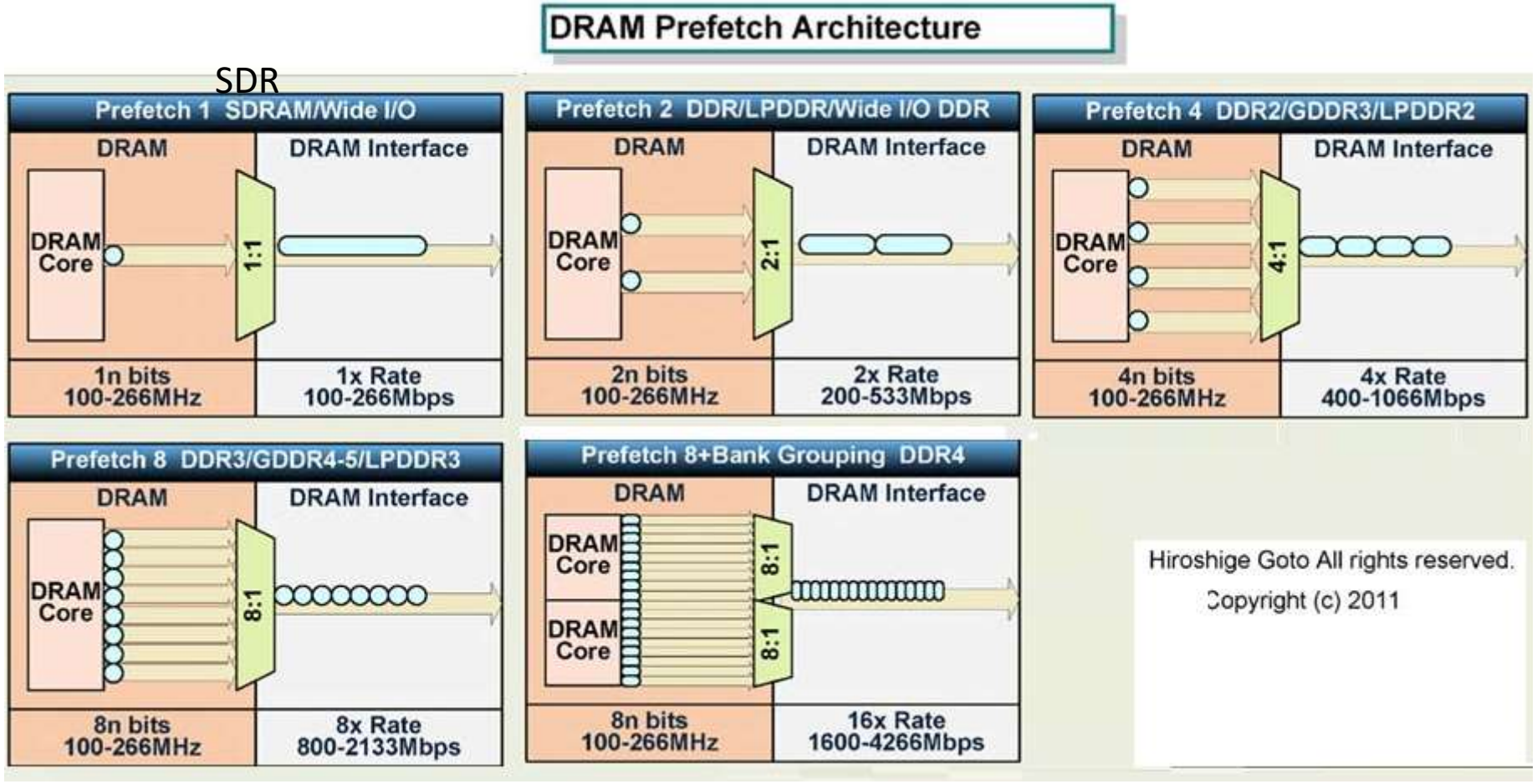


Memoria síncrona SDRAM

- DDR5 memoria más reciente: 32 datos cada T

Nombre estándar	Velocidad del reloj (MHz)	Tiempo entre señales	Velocidad del reloj de E/S (MHz)	Operaciones por segundo	Nombre del módulo	Tasa de transferencia máxima (MT/s)
DDR5-4800	600		2400 MHz	4800 millones	PC5-38400	38400 MT/s
DDR5-5200	650		2600 MHz	5200 millones	PC5-41600	41600 MT/s
DDR5-5600	700		2800 MHz	5600 millones	PC5-44800	44800 MT/s
DDR5-6000	750		3000 MHz	6000 millones	PC5-48000	48000 MT/s
DDR5-6200	775		3100 MHz	6200 millones	PC5-49600	49600 MT/s

Memoria síncrona SDRAM

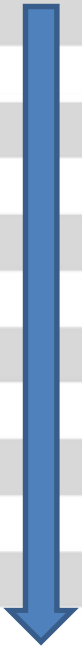


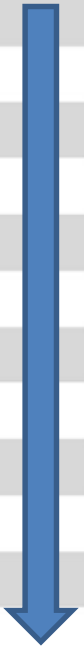
https://www.investorvillage.com/dialogs/Print.asp?msgid=11084230

Memoria SDRAM

SPEED VS. LATENCY AS MEMORY TECHNOLOGY HAS MATURED
(INDUSTRY STANDARDS)

TECHNOLOGY	MODULE SPEED (MT/s)	CLOCK CYCLE TIME (ns)	CAS LATENCY (CL)	TRUE LATENCY (ns)
SDR	100	8.00	3	24.00
SDR	133	7.50	3	22.50
DDR	335	6.00	2.5	15.00
DDR	400	5.00	3	15.00
DDR2	667	3.00	5	15.00
DDR2	800	2.50	6	15.00
DDR3	1333	1.50	9	13.50
DDR3	1600	1.25	11	13.75
DDR4	1866	1.07	13	13.93
DDR4	2133	0.94	15	14.06
DDR4	2400	0.83	17	14.17
DDR4	2666	0.75	18	13.50

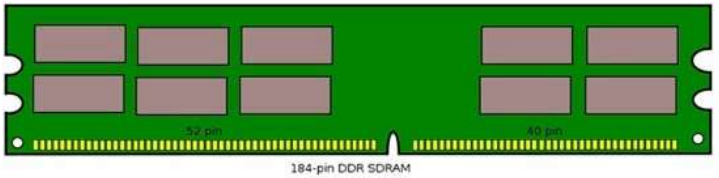




Módulos de memoria

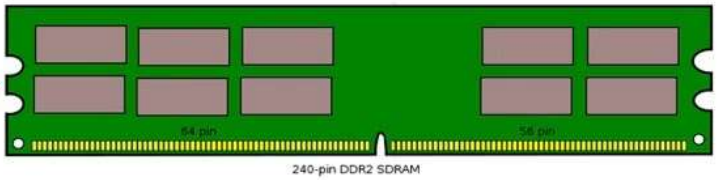
DIMM

DDR ↔



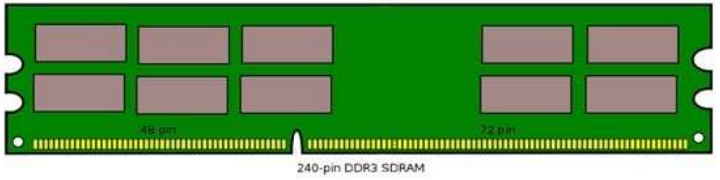
184 contactos

DDR2 ↔



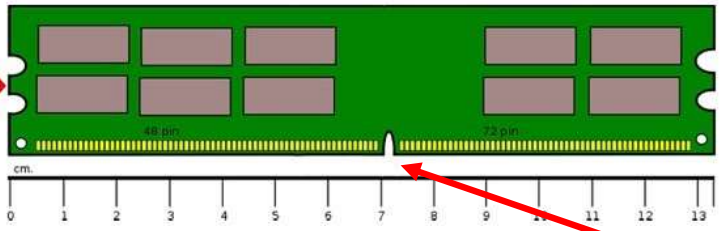
240 contactos

DDR3 ↔



240 contactos

DDR4 ↔

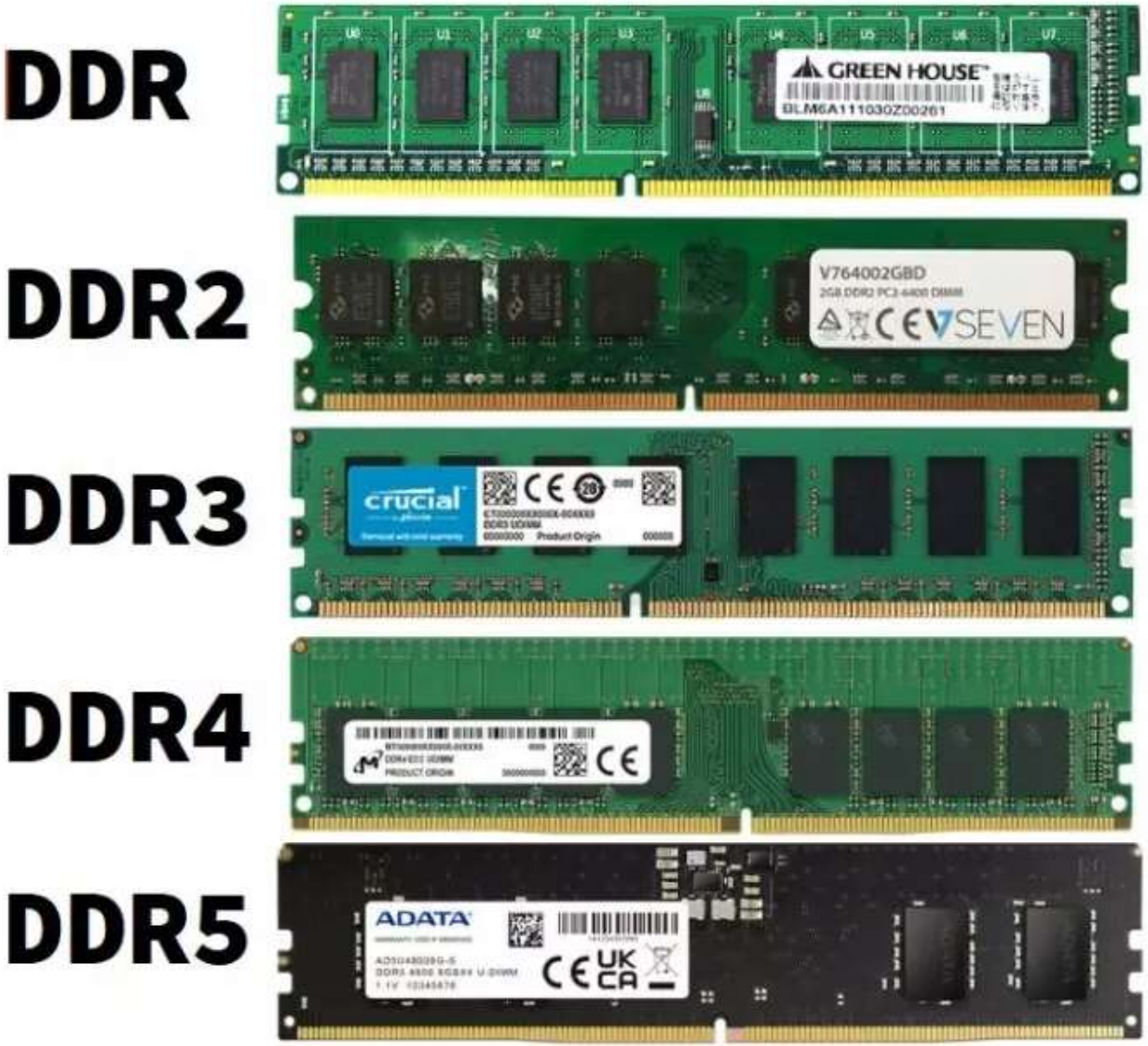


288 contactos

notch

https://linustechtips.com/main/topic/785604-difference-between-ddr1-ddr2-ddr3-ddr4-ram-in-desktop-and-laptop/

Módulos de memoria

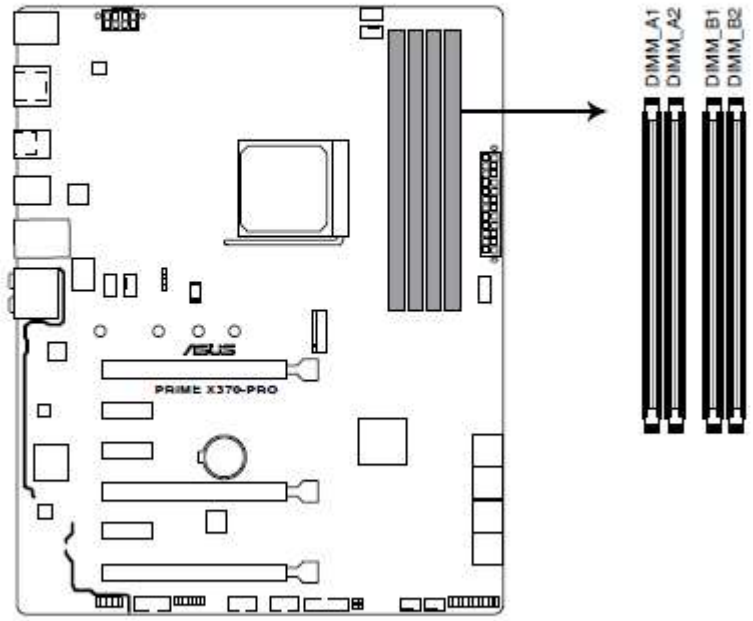


https://hardzone.es/tutoriales/componentes/diferencias-memoria-ram-ddr/

Technological advancements by the numbers, starting with DDR
2002
DDR
2.5 VOLTS
SPEED 266 MT/s
DENSITY 128Mb
133.35
31.75
28% DECREASE from DDR
50.3% INCREASE from DDR
100% INCREASE from DDR
2004
DDR2
1.8 VOLTS
SPEED 400 MT/s
DENSITY 256Mb
133.35
30.00
16.6% DECREASE from DDR2
166.5% INCREASE from DDR2
300% INCREASE from DDR2
2007
DDR3
1.5 VOLTS
SPEED 1066 MT/s
DENSITY 1Gb
133.35
30.00
20% DECREASE from DDR3
100% INCREASE from DDR3
300% INCREASE from DDR3
2014
DDR4
1.2 VOLTS
SPEED 2133 MT/s
DENSITY 4Gb
133.35
31.25

Módulos de memoria

A DDR4 module is notched differently from a DDR, DDR2, or DDR3 module. DO NOT install a DDR, DDR2, or DDR3 memory module to the DDR4 slot.



PRIME X370-PRO 288-pin DDR4 DIMM sockets

E12577_PRIME_X370-PRO_UM_v2_web_only_20170712.pdf

¿Qué memoria es más rápida?

PC3200: =DDR 400

3200 MB/s * 8 bits = 25600 Mb/s

25600 Mb/s / 64-bits = 400 MHz

400 MHz / 2 = 200 MHz base clock speed

Other memory types may use quad or even octal data rates, but if we convert those into the base clock speed, we can compare latencies. Where timings are listed in clock cycles, latency is listed in nanoseconds (ns). A CL of 2.0 sounds better than a CL of 5.0, but depending on the memory clock, it may actually be closer than we would at first expect. By converting all of the timings into nanoseconds, we can compare performance. We will save detailed comparisons for the next installment, but as an example, suppose we have two memory types - one with a CL of 4.0 and a base clock speed of 333 MHz, and the second with a CL of 2.5 and a base clock speed of 200 MHz.

CL	Clock Speed	Cycle Time	Real Latency
2.5	200 MHz	5.0 ns	12.5 ns
4.0	333 MHz	3.0 ns	12.0 ns

http://www.anandtech.com/show/1491/6

Identificación del módulo

←--------MODULO----------→ ←----------CHIP----------→

Raw Card Version	DIMM Capacity	DIMM Organization	SDRAM Density	SDRAM Organization	Number of SDRAMs	Number of Physical Ranks	Number of Banks in SDRAM	Number of Address Bits Row/Column
A	512MB	64 Meg x 64	512 Megabit	64 Meg x 8	8	1	8	13/10
	1GB	128 Meg x 64	1 Gigabit	128 Meg x 8	8	1	8	14/10
	2GB	256 Meg x 64	2 Gigabit	256 Meg x 8	8	1	8	15/10
	4GB	512 Meg x 64	4 Gigabit	512 Meg x 8	8	1	8	16/10
	8GB	1 Gig x 64	8 Gigabit	1 Gig x 8	8	1	8	16/11
B	1GB	128 Meg x 64	512 Megabit	64 Meg x 8	16	2	8	13/10
	2GB	256 Meg x 64	1 Gigabit	128 Meg x 8	16	2	8	14/10
	4GB	512 Meg x 64	2 Gigabit	256 Meg x 8	16	2	8	15/10
	8GB	1 Gig x 64	4 Gigabit	512 Meg x 8	16	2	8	16/10
	16GB	2 Gig x 64	8 Gigabit	1 Gig x 8	16	2	8	16/11
C[1]	256MB	32 Meg x 64	512 Megabit	32 Meg x 16	4	1	8	12/10
	512MB	64 Meg x 64	1 Gigabit	64 Meg x 16	4	1	8	13/10
	1GB	128 Meg x 64	2 Gigabit	128 Meg x 16	4	1	8	14/10
	2GB	256 Meg x 64	4 Gigabit	256 Meg x 16	4	1	8	15/10

Identificación del módulo

Memory configurations

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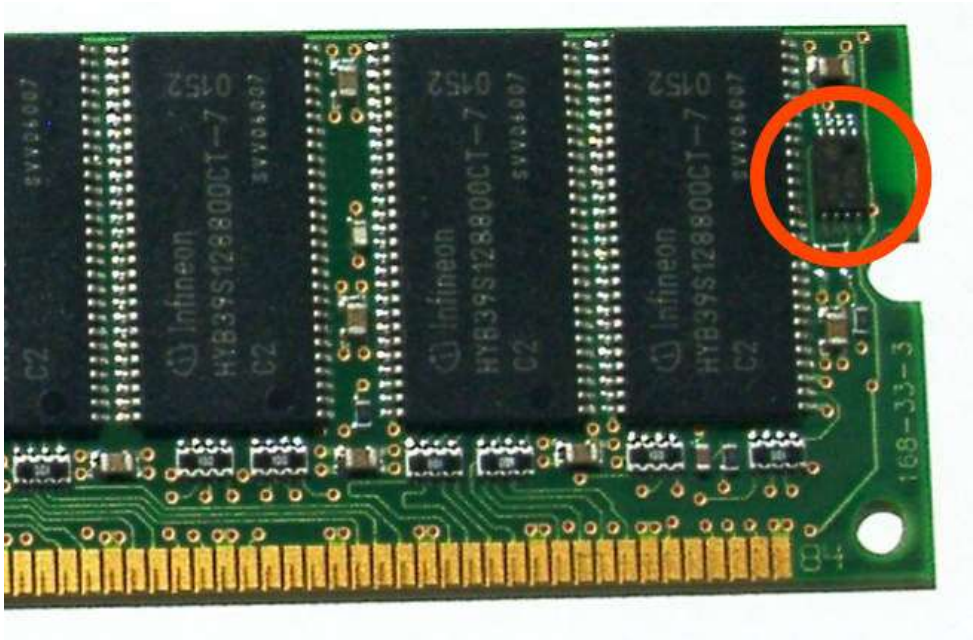
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- Always install the DIMMS with the same CAS Latency. For an optimum compatibility, we recommend that you install memory modules of the same version or data code (D/C) from the same vendor. Check with the vendor to get the correct memory modules.
- Visit the ASUS website for the latest QVL.

Identificación del módulo

- Estándar para acceder a las características de un módulo de memoria
- http://www.simmtester.com/page/news/showpubnews.asp?num=153
- https://en.wikipedia.org/wiki/Serial_presence_detect



Identificación del módulo

- **Serial presence detect** (**SPD**): **Pequeña memoria de 256 Bytes que describe el módulo de memoria.**
- Standardized way to automatically access information about a computer memory module. When an ordinary modern computer is turned on, it starts by doing a power-onself-test (POST). Since about the mid-1990s, this process includes automatically configuring the hardware currently present. SPD is a memory hardware feature that makes it possible for the computer to know what memory is present, and what timings to use to access the memory.
- Some computers adapt to hardware changes completely automatically. In most cases, there is a special optional procedure for accessing BIOS parameters, to view and potentially make changes in settings. It may be possible to control how the computer uses the memory SPD data—to choose which settings to use, to selectively modify the memory timings, or possibly to completely over-ride the SPD data

The following table is intended to be an **example** of the SPD data for a PC3-8500, double-sided 2GB (2 x 128 Meg x 64) non-ECC, 240-pin unbuffered DDR3 SDRAM DIMM. This example shows a module comprised of sixteen 128 Meg x 8 DRAM components.

Table 7.2.1 — Serial Presence Detect, Example Raw Card Version B (Part 1 of 2)

Byte Number	Byte Attribute Description	SPD Entry Value	Hexadecimal Byte Value
0	CRC range, EEPROM bytes, bytes used	Bytes 0–116/256/176	92
1	SPD revision	Rev 0.0	00
2	DRAM device type	DDR3 SDRAM	0B
3	Module type (form factor)	UDIMM	02
4	SDRAM Device density and banks	1Gb, 8 banks	02
5	SDRAM device row and column count	14 rows, 10 columns	11
6	Byte 6 reserved	None	00
7	Module ranks and device DQ count	2 Ranks, x8 DQs	09
8	ECC tag and module memory Bus width	Non-ECC, 64-bit Bus	03
9	Fine timebase dividend/divisor (in pico seconds)	5/2 = 2.5ps	52
10	Medium timebase dividend	1ns	01
11	Medium timebase divisor	8ns	08
12	Minimum SDRAM cycle time ($^{t}CK_{MIN}$)	1.875ns	0F
13	Byte 13 reserved	None	00
14	CAS latencies supported (CL4 ≥ CL11)	CAS 8, 7, and 6	1C
15	CAS latencies supported (CL12 ≥ CL18)	None	00
16	Minimum CAS latency time ($^{t}AA_{MIN}$)	13.125ns	69
17	Minimum write recovery time ($^{t}WR_{MIN}$)	15ns	78
18	Minimum $\overline{RAS}$-to-$\overline{CAS}$ delay ($^{t}RCD_{MIN}$)	13.125ns	69
19	Minimum row ACTIVE-to-ROW ACTIVE delay ($^{t}RRD_{MIN}$)	7.5ns	3C
20	Minimum row PRECHARGE delay ($^{t}RP_{MIN}$)	13.125ns	69
21	Upper nibble for ^{t}RAS and ^{t}RC	1(2C) and 1(A4)	11
22	Minimum ACTIVE-to-PRECHARGE delay ($^{t}RAS_{MIN}$)	0x12C = 37.5ns	2C

Table 7.2.1 — Serial Presence Detect, Example Raw Card Version B (Part 2 of 2)

Byte Number	Byte Attribute Description	SPD Entry Value	Hexadecimal Byte Value
23	Minimum ACTIVE-to-ACTIVE/REFRESH delay ($^tRC_{MIN}$)	0x1A4 = 52.5ns	A4
24	Minimum refresh recovery delay ($^tRFC_{MIN}$) LSB	(combo bytes 24, 25)	70
25	Minimum refresh recovery delay ($^tRFC_{MIN}$) MSB	0x370=110ns $^tRFC_{MIN}$	03
26	Minimum internal WRITE-TO-READ command delay ($^tWTR_{MIN}$)	7.5ns	3C
27	Minimum internal READ-to-PRECHARGE command delay ($^tRTP_{MIN}$)	7.5ns	3C
28	Minimum four active window delay ($^tFAW_{MIN}$) LSB	(combo bytes 28, 29)	01
29	Minimum four active window delay ($^tFAW_{MIN}$) LSB	0x12C = 37.5ns	2C
30	SDRAM device output drivers supported	RZQ/7 supported	02
31	SDRAM device thermal and refresh options	ASR, extended temperature	05
32–59	Bytes 32–59 reserved	None	00000000000000000000
60	Module height (NOMINAL)	30 mm	0F
61	Module thickness (MAX)	DR, 4 mm MAX	11
62	Raw card ID reference	Raw Card B	01
63	DRAM address mapping edge connector	Standard	00
64–116	Bytes 64–116 reserved	None	00000000000000000000
117	Module manufacturer ID (LSB)	Manufacturer specific	80
118	Module manufacturer ID (MSB)	Manufacturer specific	2C
119	Module manufacturer location ID	Variable data	00
120	Module manufacturer year	Variable data	00
121	Module manufacturer week	Variable data	00
122–125	Module serial number	Variable data	00000000
126–127	CRC	Calculated value	027B
128–145	Module part number	ASCII values	16JTF25664AY-1G1A1
146–147	Module revision code	Variable data	41
148	DRAM Device manufacturer ID (LSB)	Manufacturer specific	80
149	DRAM device manufacturer (MSB)	Manufacturer specific	2C
150–175	Manufacturer-reserved bytes 150–175	Reserved	00000000000000000000
176–255	Customer-reserved bytes 176–255	Reserved	FFFFFFFFFFFFFFFFFFFF

Identificación del módulo

- ® Extreme Memory Profile
- Los datos en la SPD son muy conservadores, para así garantizar funcionamiento correcto en todos los sistemas
- XMP (Extreme Memory Profile) añade 2 perfiles:
  - XMP profile 1, ‘Enthusiast / Certified’ settings
  - XMP profile 2 ‘Extreme’ settings
- Usa los bytes 176–255 de la SPD

https://www.intel.com/content/www/us/en/gaming/extreme-memory-profile-xmp.html

Identificación del módulo

Intel® XMP Profiles

Intel® Extreme Memory Profiles are traditionally designed with two distinct performance profiles:

- **Profile 1** is used for the Enthusiast / Certified settings and is the profile that is tested under the Intel® Extreme Memory Certification program.
- **Profile 2** is designed to host the Extreme or Fastest possible settings that have no guard band and may or may not work on every system.

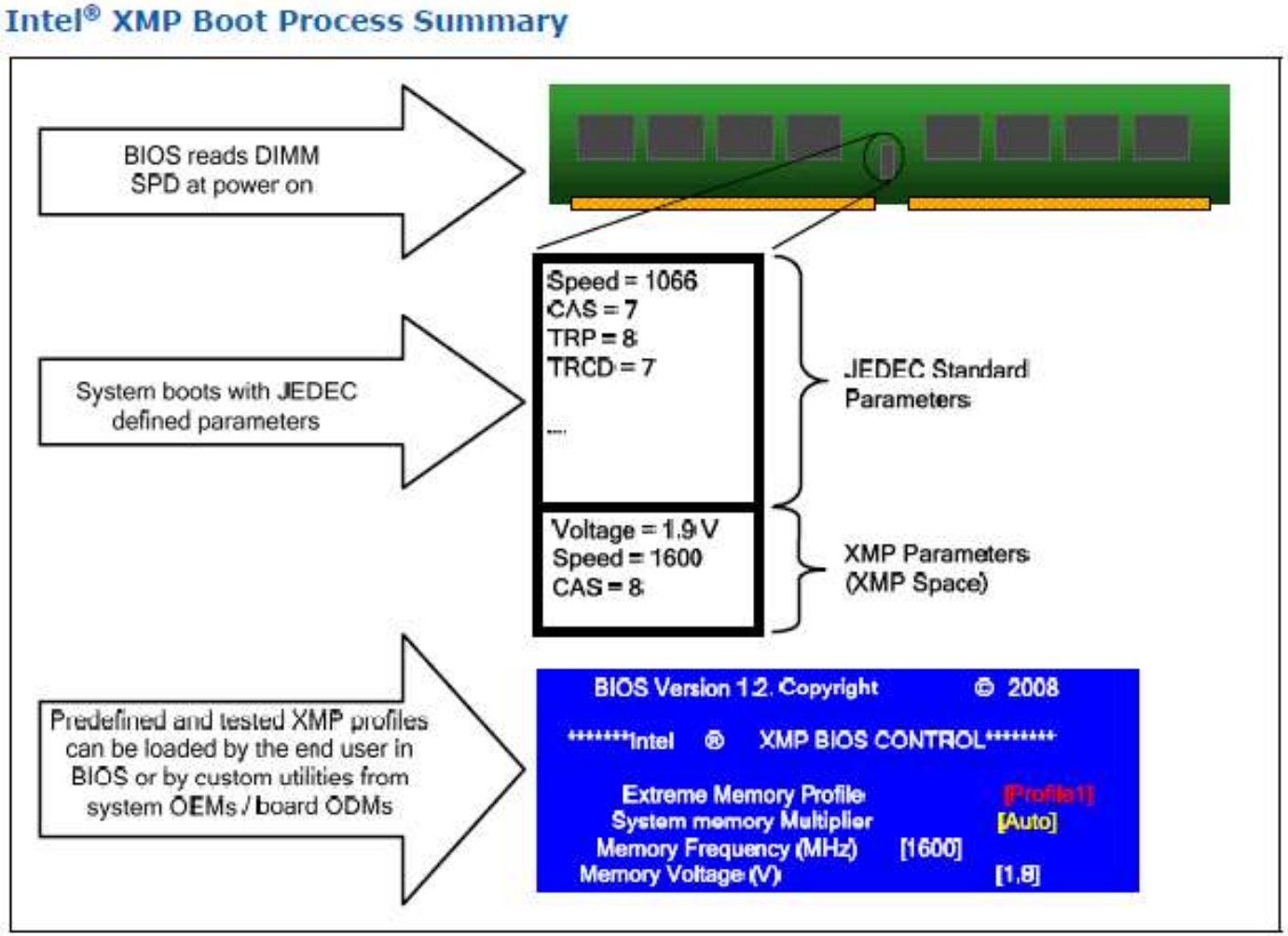
It should also be noted that Extreme Memory Profiles are not always defined as over-frequency / over-voltaged parts. In some cases, Extreme Memory Profiles can be used to define extremely power savvy settings or extremely fast latencies.

As Mentioned in Section 3.2, XMP DIMMs use bytes 176–255 to program XMP specific details. The following is the breakdown of the XMP SPD space.

- **Bytes 176 -185**: Global Parameters for Intel Extreme Memory used across all profiles
- **Bytes 186 – 218**: Profile 1 programming
- **Bytes 219 – 254**: Profile 2 programming

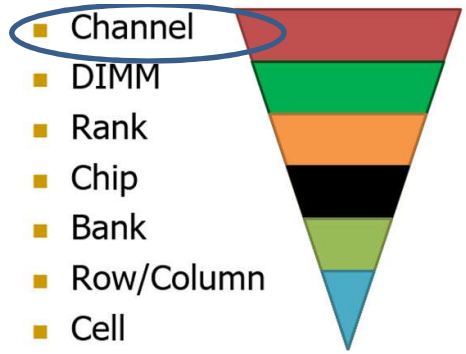
https://www.intel.com/content/dam/doc/white-paper/extreme-memory-profile-ddr3-technology-paper.pdf

Identificación del módulo



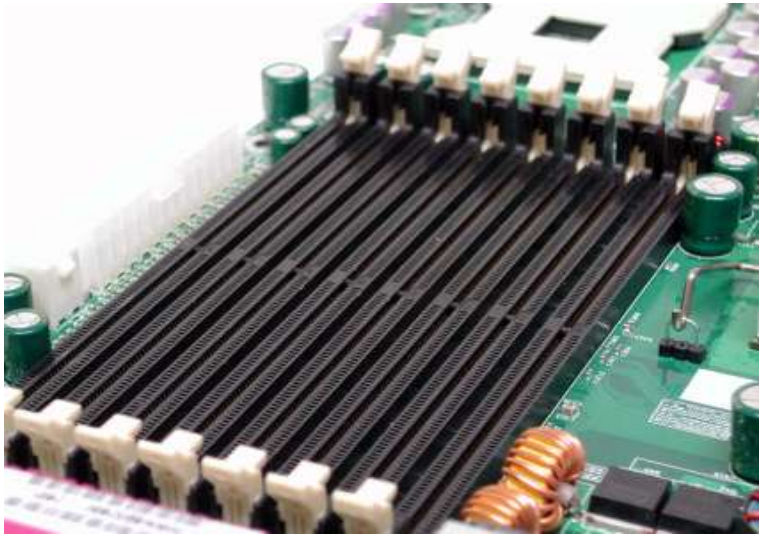
https://www.intel.com/content/dam/doc/white-paper/extreme-memory-profile-ddr3-technology-paper.pdf

Channel



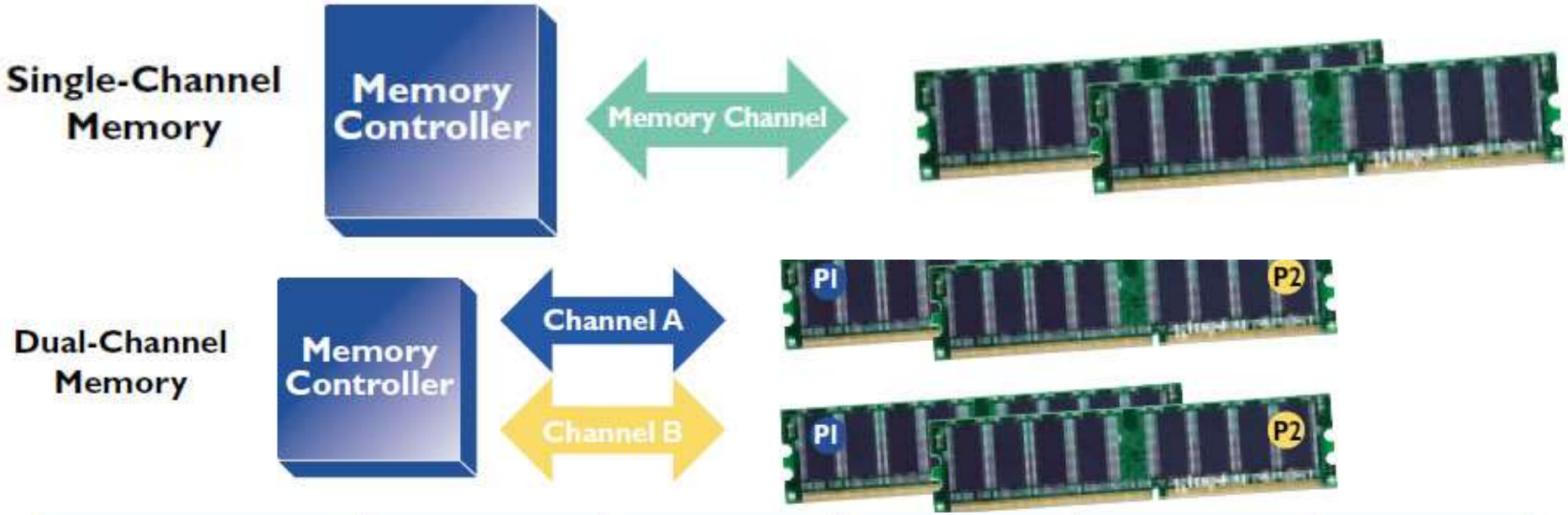
- Un canal transmite 64 bits, está formado por uno o más DIMMS. Sólo se accede a un DIMM a la vez.





Channel

- Una computadora puede tener varios canales



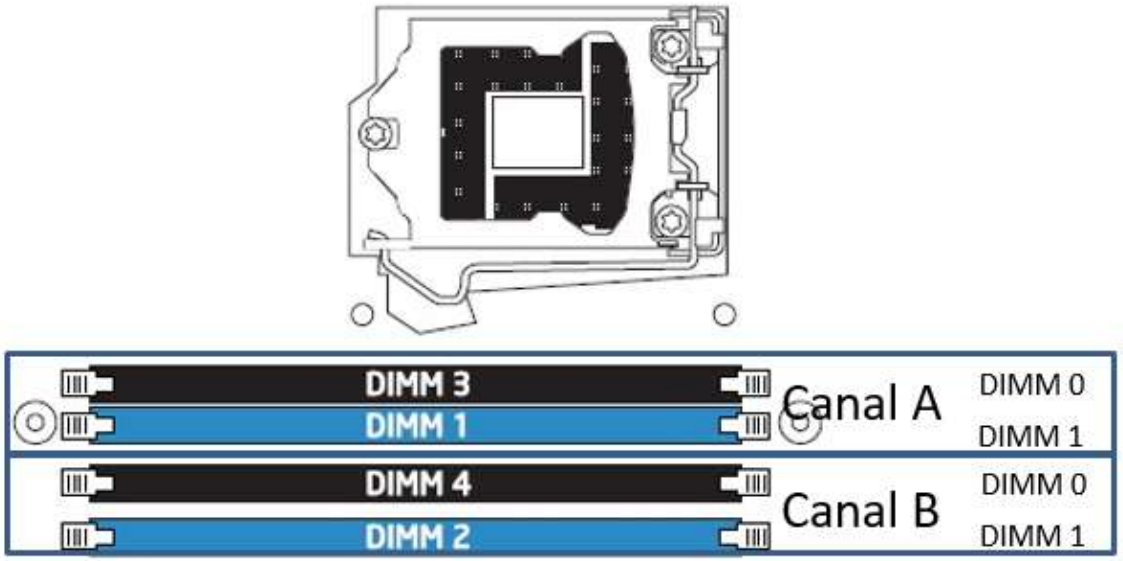
PEAK BANDWIDTH	DATA BITS ACCESSED	PC-133	PC2100 DDR266	PC2700 DDR333	PC3200 DDR400
Single-Channel	64	1.1GB/s	2.1GB/s	2.7GB/s	3.2GB/s
Dual-Channel	128		4.2GB/s	5.4GB/s	6.4GB/s

Channel

- Single Channel: Todos los módulos de memoria intercambian información a través de un único canal, 64 bits, para ello sólo es necesario colocar los módulos DIMM en los conectores de ese canal.
- Dual Channel: Existen 2 canales de memoria, 64 + 64 bits, cada canal puede tener varios conectores DIMM. Se reparten los módulos de memoria entre los 2 canales. Pueden intercambiarse datos a través de 2 canales simultáneamente.

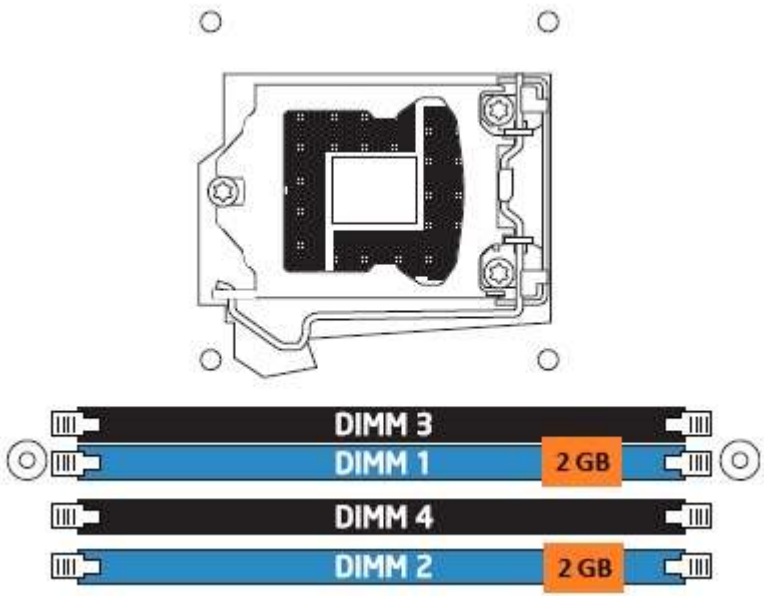
Channel

- Código de colores en los conectores DIMM de la placa base.
- Los conectores de igual color están todos en diferente canal.
- Los módulos que se instalen deben montarse y completar los conectores del mismo color.



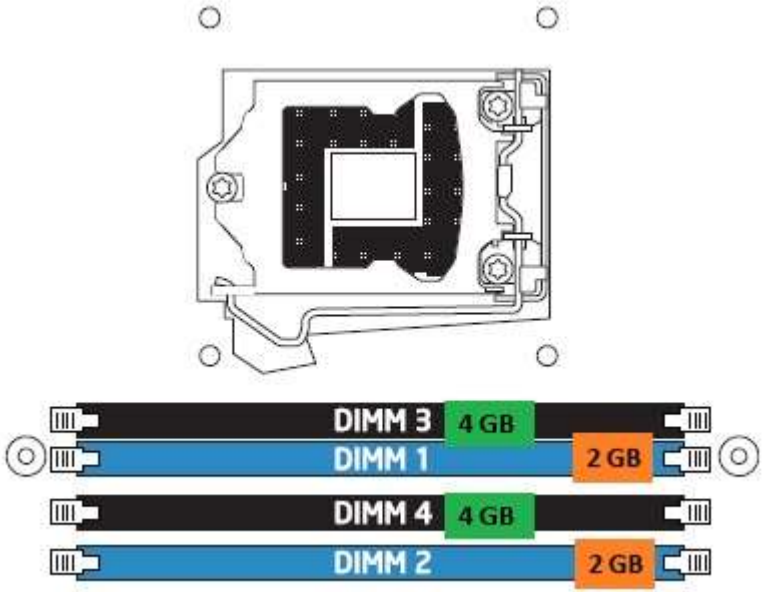
Channel

- Placa Dual Channel trabajando en modo Dual Channel



Canal dual con 2 DIMM

Canal A
Canal B



Canal dual con 4 DIMM

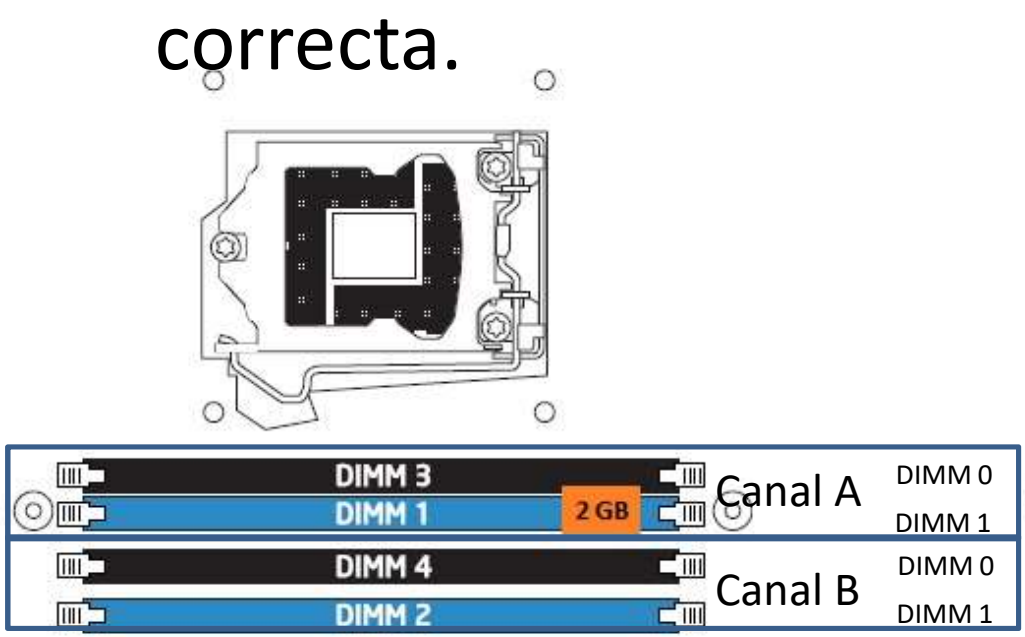
Channel

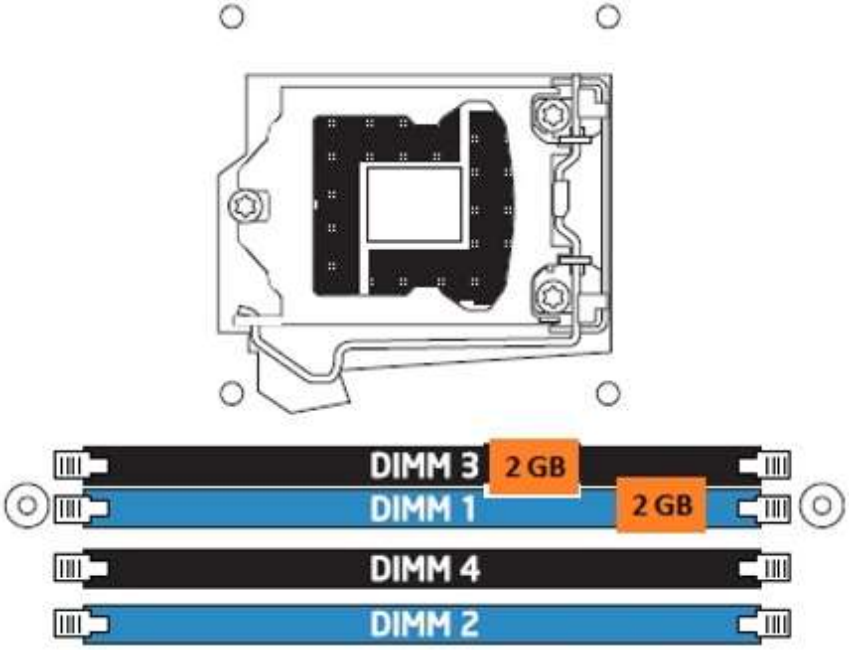
- To achieve dual channel mode, following conditions must be met:
  - Matched DIMM configuration in each channel
  - Same density (128 MB, 256 MB, 512 MB, etc.)
  - Matched in both Channel A and Channel B memory channels
  - Populate symmetrical memory slots (Slot 0 or Slot 1)
  - Configurations that do not match the above conditions revert to Single Channel mode.
- The following conditions do **not** need to be met:
  - Same brand
  - Same timing specifications
  - Same DDR speed
  - Memory channel speed is decided by the slowest DIMM module populated in the system.

http://www.intel.com/support/motherboards/desktop/sb/cs-011965.htm

Channel

- Una placa Dual Channel trabaja en modo Single Channel cuando la configuración no es correcta.





https://www.intel.es/content/www/es/es/support/articles/000005657/boards-and-kits.html

Channel

- Al arrancar, se detecta la configuración de la memoria y es posible este mensaje de alerta:
  - Alerta: el máximo rendimiento de memoria se consigue con la misma cantidad de memoria instalada en cada canal. Pulse cualquier tecla para continuar..
- Con los módulos DIMM actualmente instalados, el equipo está configurado en modo Single Channel, pero puede ajustarse al modo Dual Channel. Si apaga y reorganiza correctamente los DIMM, puede establecer el modo de canal dual.

Channel

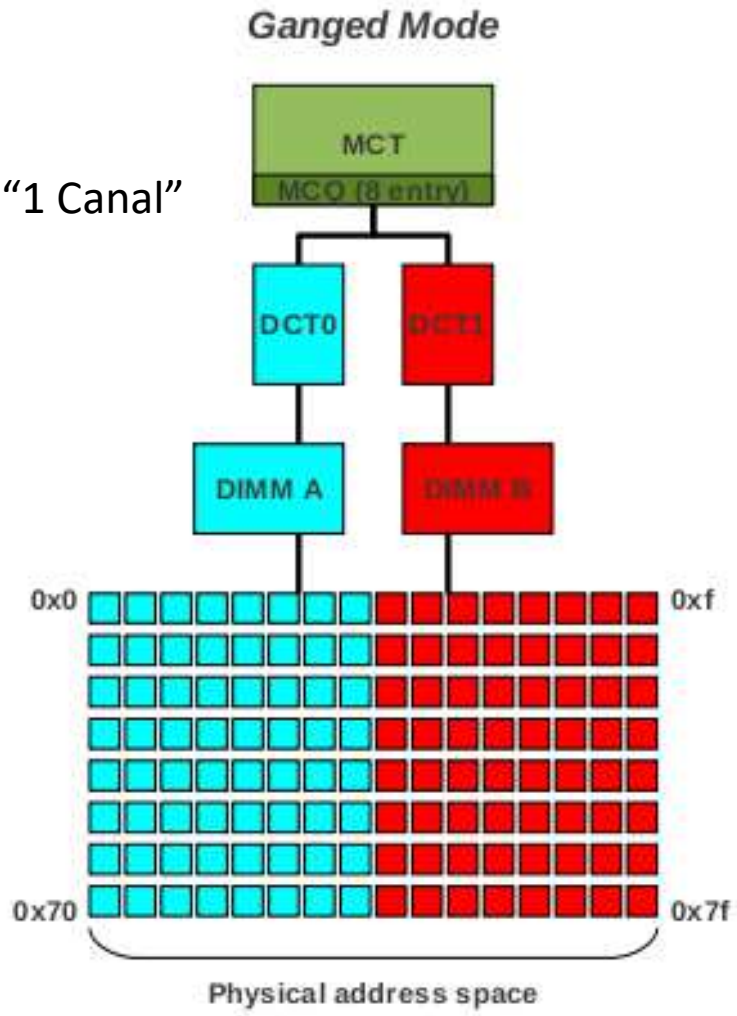
- Con Dual Channel no tenemos una mejora global de rendimiento del 100%.
- La memoria principal DIMM sólo es una parte del sistema de memoria de una computadora.
- La memoria caché también es parte de ese sistema.
- Pizarra: 90% accesos a caché y 10% accesos a MP. Mejora con Dual Channel

http://www.utiltecnico.com/modos-de-doble-canal-en-memorias-ram-dual-channel-flex-mode-y-single-channel/

http://www.ilsistemista.net/index.php/hardware-analysis/3-the-phenom-phenomii-memory-controller-and-the-ganged-vs-unganged-question.html?start=1

Channel

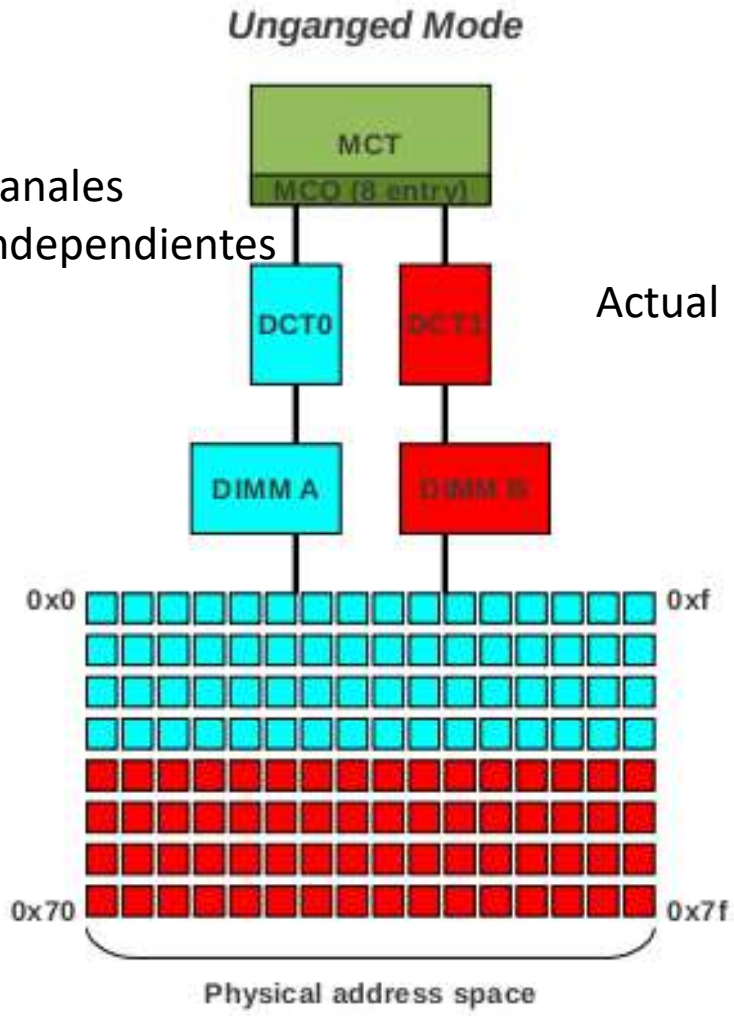
"1 Canal"



Se leen 128 bytes consecutivos a partir de una dirección de memoria

Canales independientes

Actual



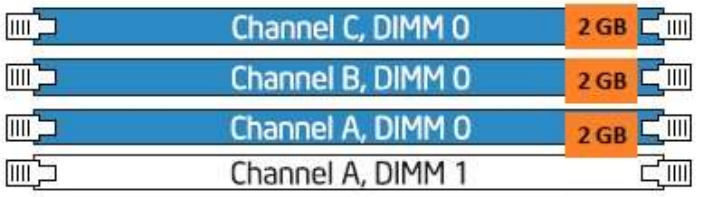
Se leen 64 bytes consecutivos en una dirección + 64 Bytes consecutivos de otra dirección

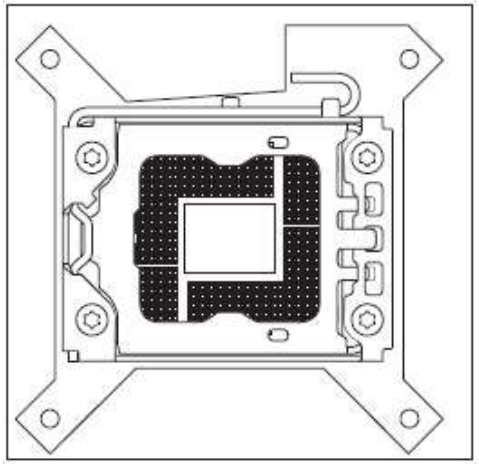
Channel

- Ganged Mode (modo agrupado)
  - Los dos canales forman un canal lógico de 128 bits
  - El controlador accede a ambos módulos al mismo tiempo como si fueran uno solo.
  - Ideal para cargas de trabajo de un solo hilo que requieren acceso ancho y secuencial a memoria.
  - Más ancho de banda, pero menos concurrencia.
- Unganged Mode (modo desagrupado)
  - Los dos canales son independientes, 64 bits cada uno.
  - Permite accesos independientes y concurrentes desde múltiples hilos o núcleos del procesador.
  - Ideal para cargas multihilo y sistemas multitarea.
  - Puede mejorar el rendimiento general en escenarios modernos.

Channel

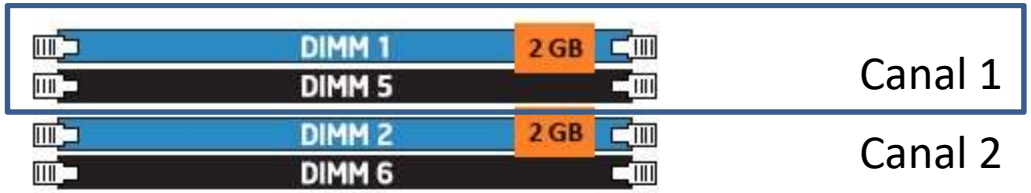
- Triple Channel: 3 canales de dan varios modos posibles:
  - El Triple Channel se activa cuando se instalan módulos idénticos en cada uno de los tres conectores azules.
  - Si sólo dos de los conectores azules tienen módulos DIMMs, se activa Dual Channel.
  - Con un módulo o 2 en el canal A tenemos Single Channel





Channel

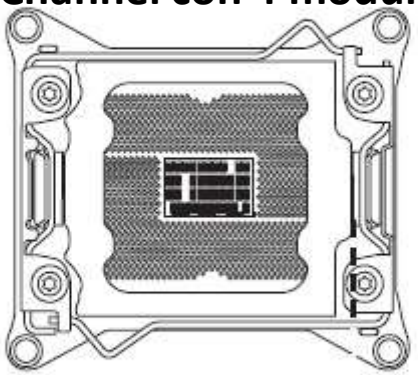
- Quad Channel: cuando existen 4 canales y 4 módulos DIMM (o múltiplo de 4) idénticos en capacidad y velocidad instalados convenientemente. Cuando se instalan 2 módulos, el sistema funciona en modo dual. Cuando se instalan 3 módulos, el sistema funciona en modo triple.



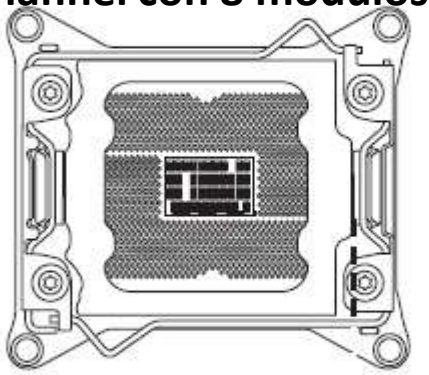
Quad-Channel con 4 módulos DIMMs



Quad-Channel con 8 módulos DIMMs



Azul: primer DIMM de cada canal
Negro: segundo DIMM de cada canal

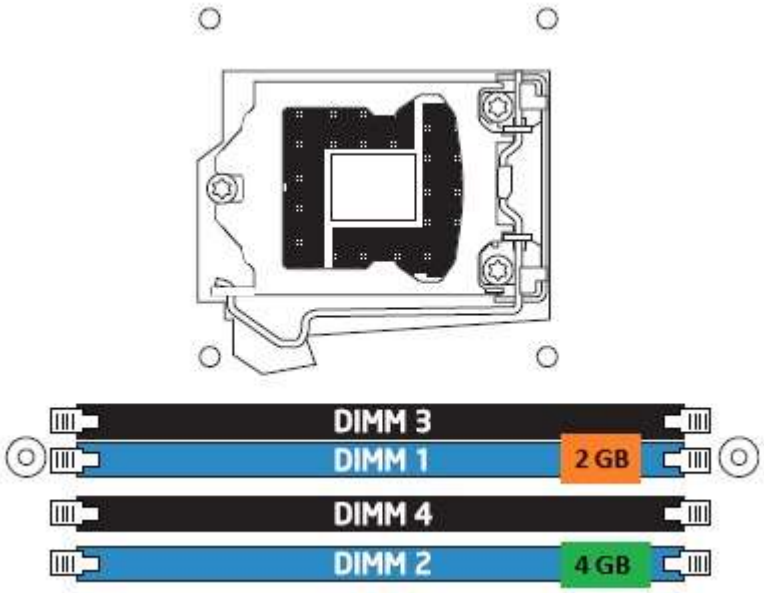






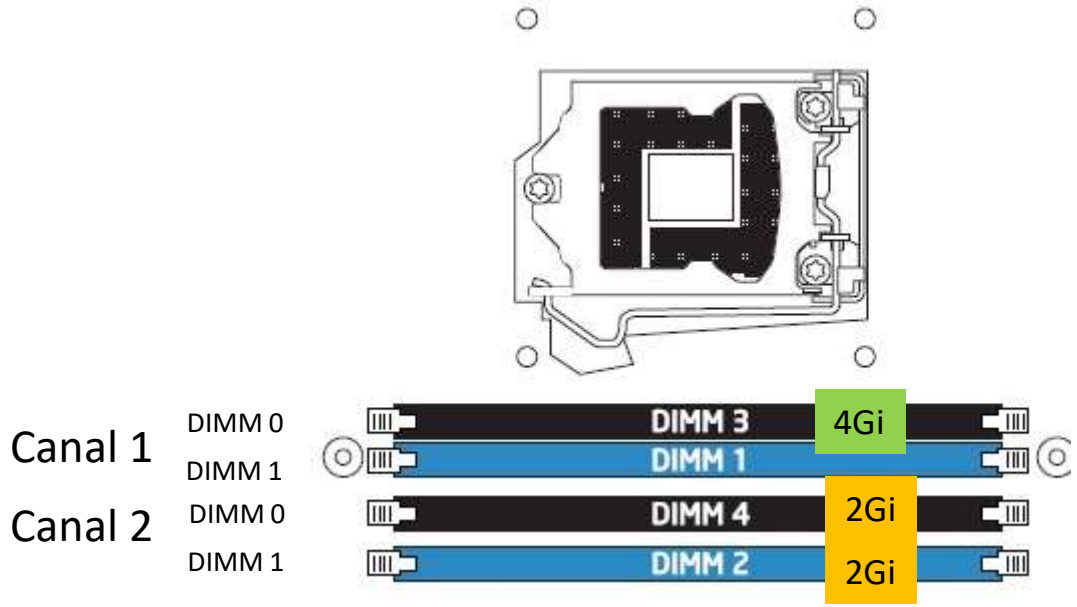
Channel

- Modo FLEX: Single Channel y Dual Channel flexible
  - El DIMM de 2 GiB en la ranura 1 y los 2 GiB inferiores del DIMM en la ranura 2 funcionan juntos en el modo dual.
  - El resto (superior) de los 2 GiB de la DIMM en la ranura 2 funciona en modo single.



Channel

- Modo FLEX
  - En canal 1, un módulo de 4 Gi
  - En canal 2, dos módulos de 2 Gi



Channel

Memory configurations

You may install 2 GB, 4 GB, 8 GB, 16 GB, and 32 GB unbuffered and non-ECC DDR4 DIMMs into the DIMM sockets.

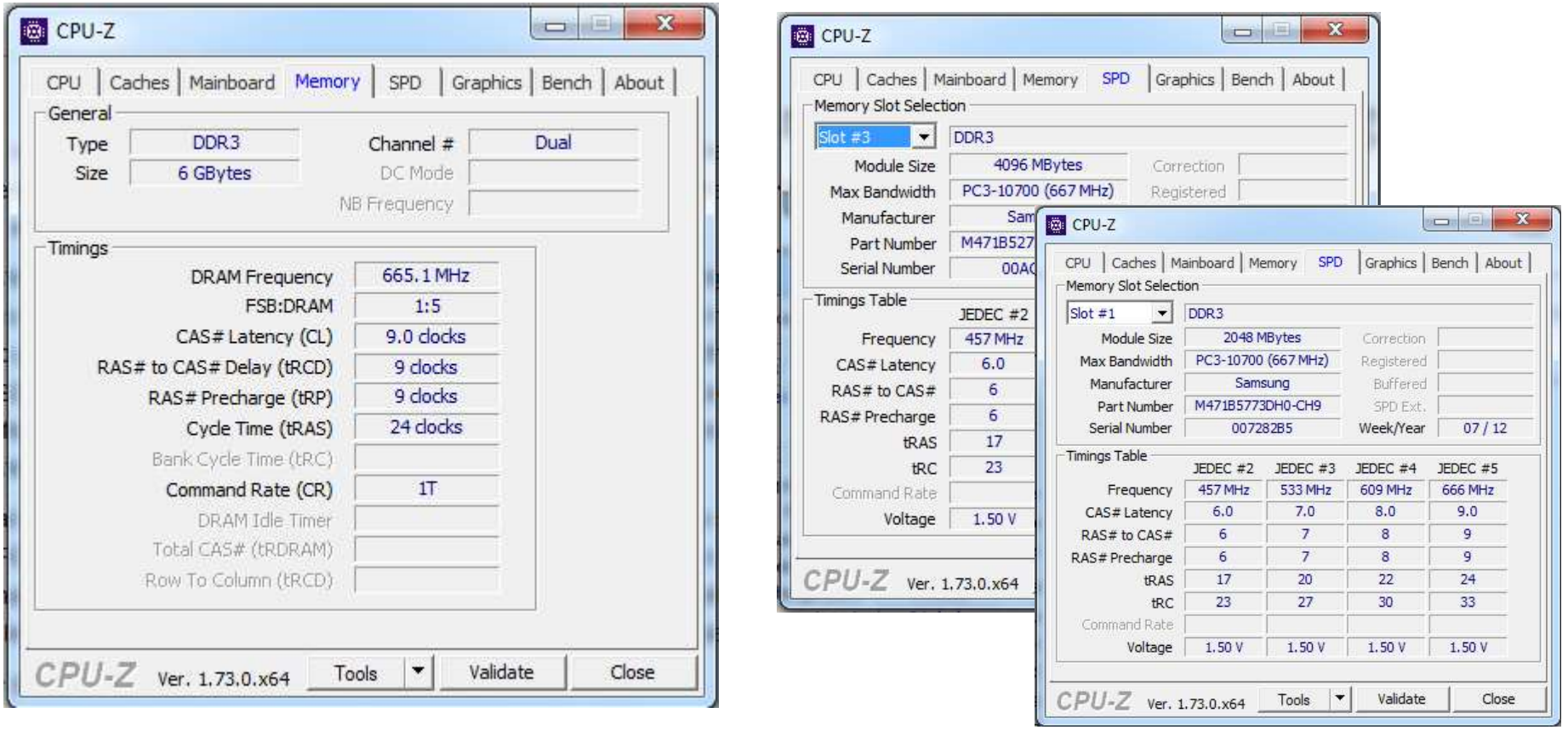
MODO FLEX

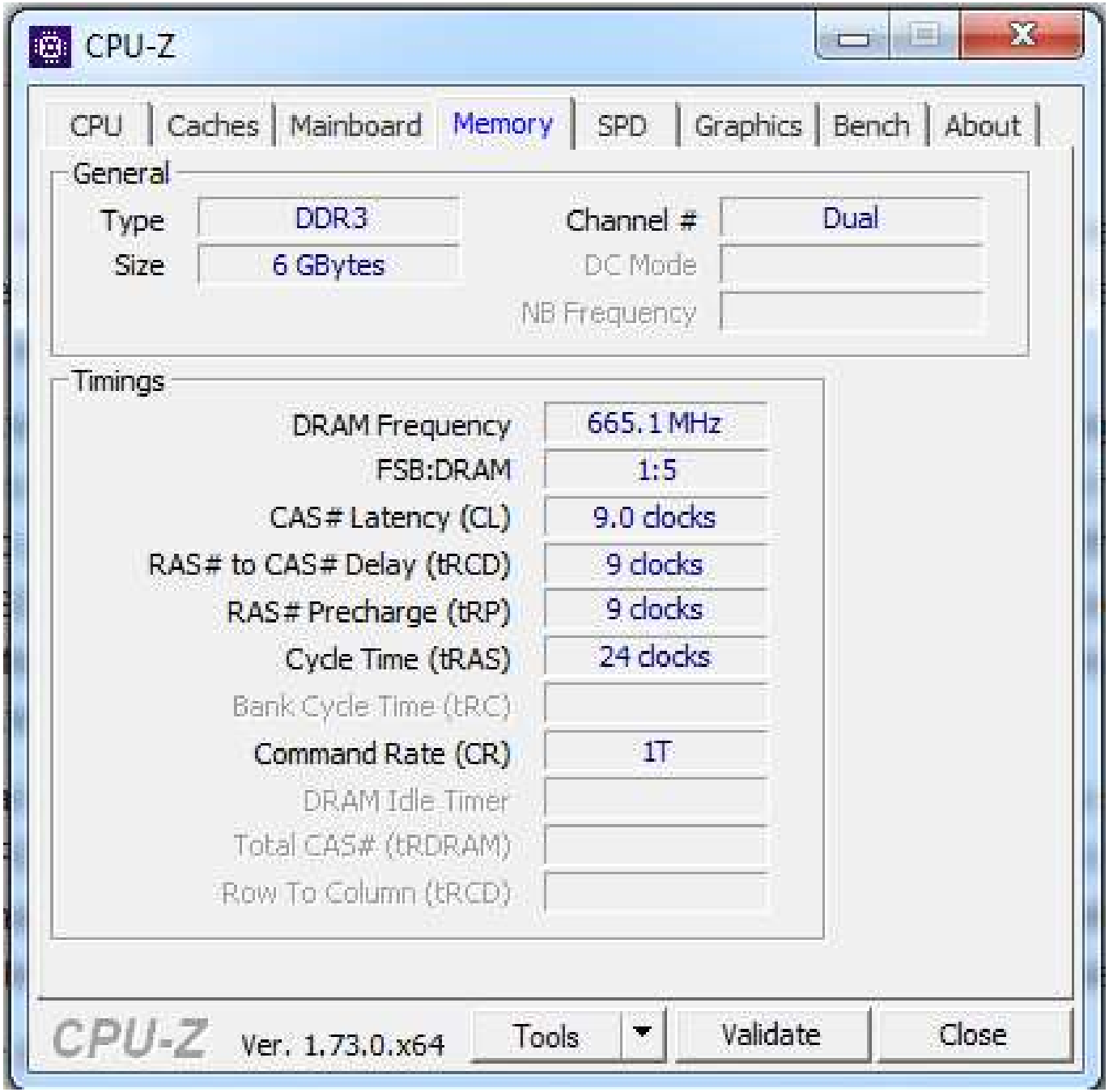
You may install varying memory sizes in Channel A and Channel B. The system maps the total size of the lower-sized channel for the dual-channel configuration. Any excess memory from the higher-sized channel is then mapped for single-channel operation.

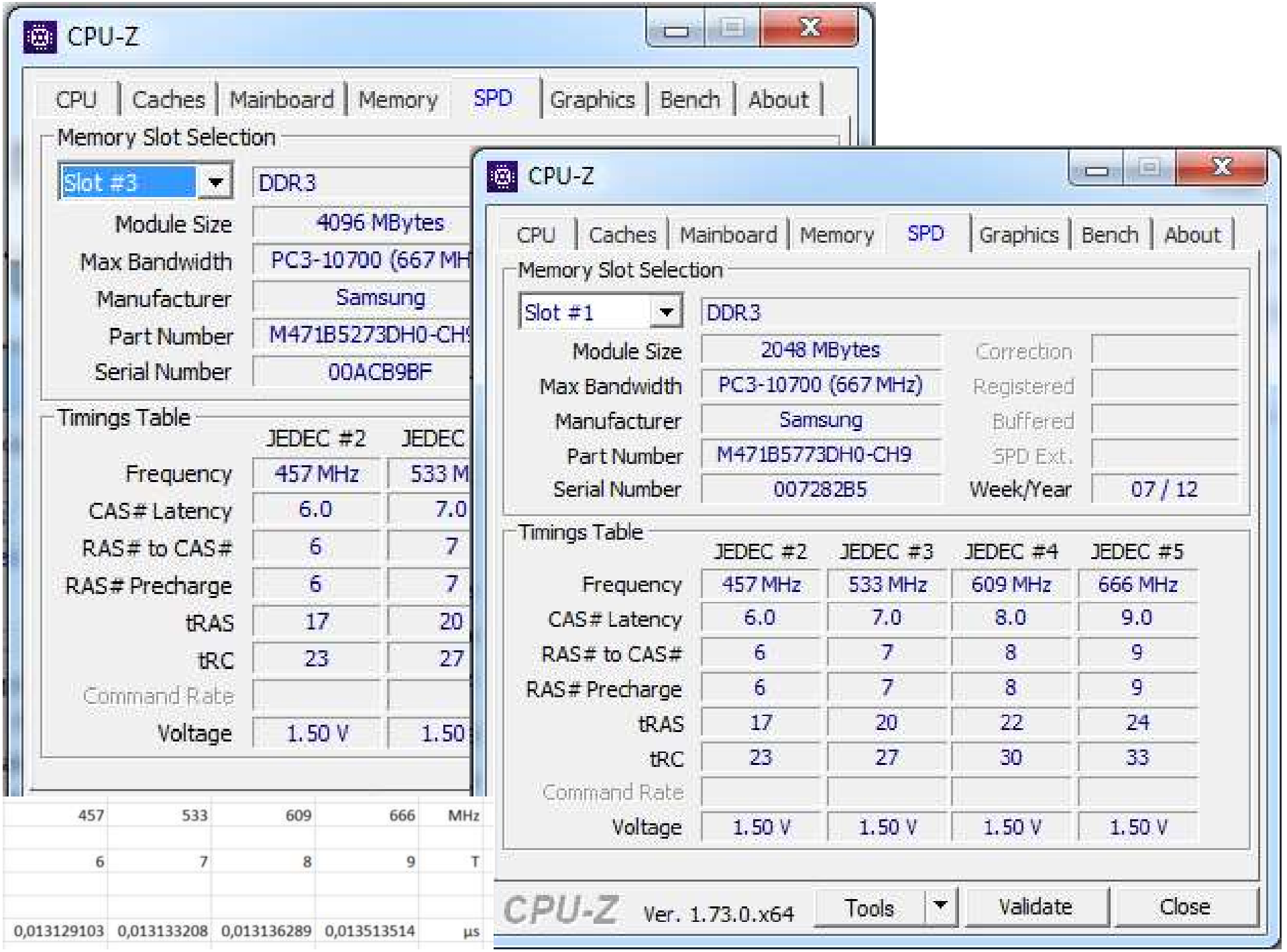
- The default memory operation frequency is dependent on its Serial Presence Detect (SPD), which is the standard way of accessing information from a memory module. Under the default state, some memory modules for overclocking may operate at a lower frequency than the vendor-marked value.
- For system stability, use a more efficient memory cooling system to support a full memory load or overclocking condition.
- Always install the DIMMS with the same CAS Latency. For an optimum compatibility, we recommend that you install memory modules of the same version or data code (D/C) from the same vendor. Check with the vendor to get the correct memory modules.
- Visit the ASUS website for the latest QVL.

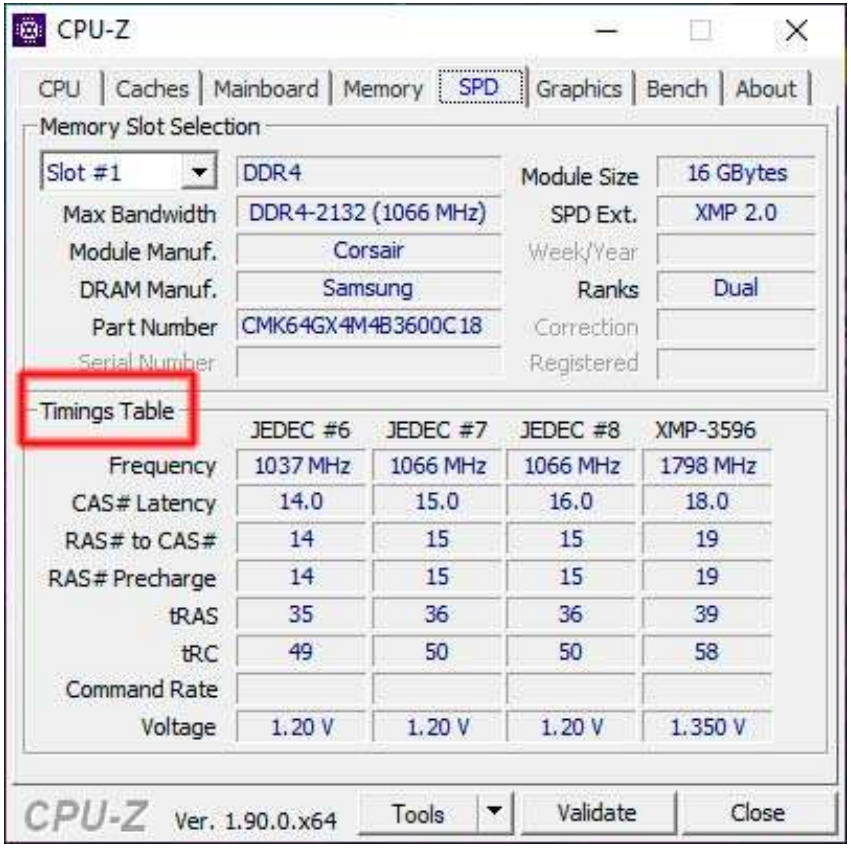
CPU-Z

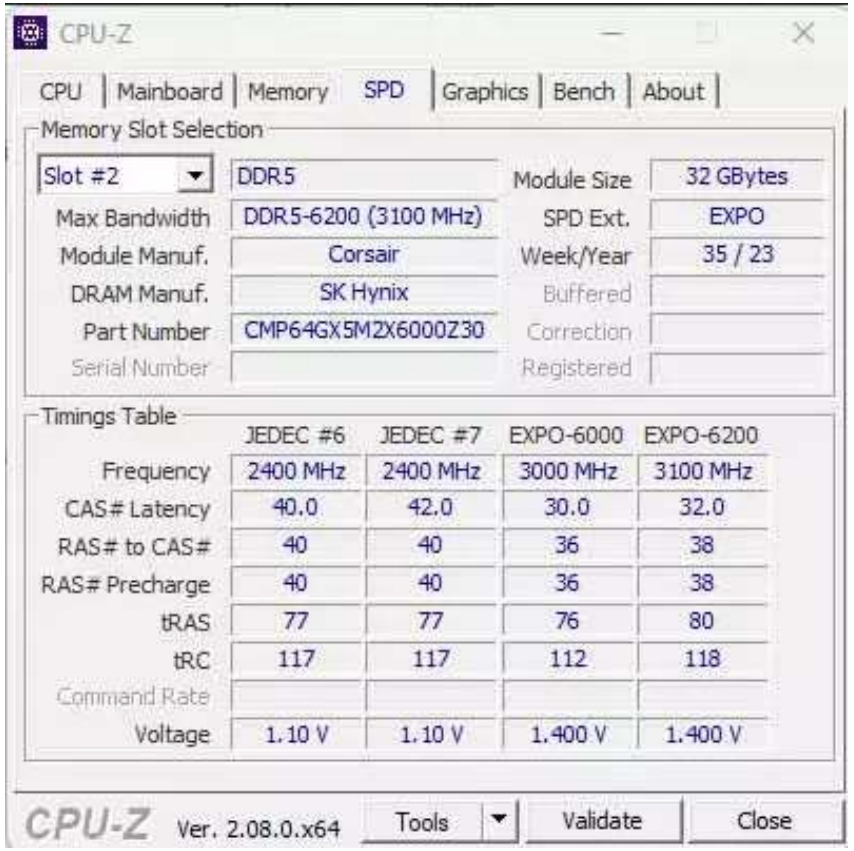
- Herramienta gratuita de diagnóstico que muestra información del PC




CPU-Z
CPU
Caches
Mainboard
Memory
SPD
Graphics
Bench
About
General
Type
DDR3
Size
6 GBytes
Channel #
Dual
DC Mode
NB Frequency
Timings
DRAM Frequency
665.1 MHz
FSB:DRAM
1:5
CAS# Latency (CL)
9.0 clocks
RAS# to CAS# Delay (tRCD)
9 clocks
RAS# Precharge (tRP)
9 clocks
Cycle Time (tRAS)
24 clocks
Bank Cycle Time (tRC)
Command Rate (CR)
1T
DRAM Idle Timer
Total CAS# (tRDRAM)
Row To Column (tRCD)
CPU-Z
Ver. 1.73.0.x64
Tools
Validate
Close


CPU-Z
CPU | Caches | Mainboard | Memory | SPD | Graphics | Bench | About
Memory Slot Selection
Slot #3
DDR3
Module Size 4096 MBytes
Max Bandwidth PC3-10700 (667 MH
Manufacturer Samsung
Part Number M471B5273DH0-CH
Serial Number 00ACB9BF
Timings Table
JEDEC #2 JEDEC
Frequency 457 MHz 533 M
CAS# Latency 6.0 7.0
RAS# to CAS# 6 7
RAS# Precharge 6 7
tRAS 17 20
tRC 23 27
Command Rate
Voltage 1.50 V 1.50
457 533 609 666 MHz
6 7 8 9 T
0,013129103 0,013133208 0,013136289 0,013513514 µs
CPU-Z
CPU | Caches | Mainboard | Memory | SPD | Graphics | Bench | About
Memory Slot Selection
Slot #1
DDR3
Module Size 2048 MBytes
Max Bandwidth PC3-10700 (667 MHz)
Manufacturer Samsung
Part Number M471B5773DH0-CH9
Serial Number 007282B5
Correction
Registered
Buffered
SPD Ext.
Week/Year 07 / 12
Timings Table
JEDEC #2 JEDEC #3 JEDEC #4 JEDEC #5
Frequency 457 MHz 533 MHz 609 MHz 666 MHz
CAS# Latency 6.0 7.0 8.0 9.0
RAS# to CAS# 6 7 8 9
RAS# Precharge 6 7 8 9
tRAS 17 20 22 24
tRC 23 27 30 33
Command Rate
Voltage 1.50 V 1.50 V 1.50 V 1.50 V
CPU-Z Ver. 1.73.0.x64
Tools
Validate
Close

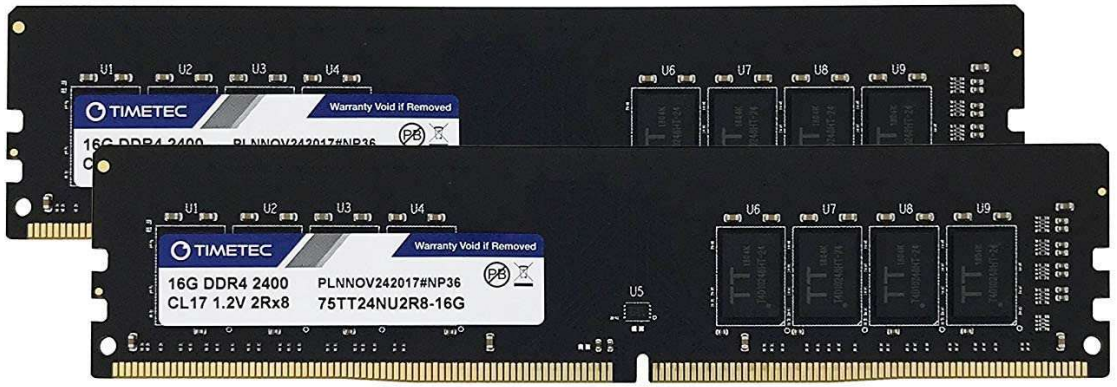

CPU-Z
CPU
Caches
Mainboard
Memory
SPD
Graphics
Bench
About
Memory Slot Selection
Slot #1
DDR4
Module Size
16 GBytes
Max Bandwidth
DDR4-2132 (1066 MHz)
SPD Ext.
XMP 2.0
Module Manuf.
Corsair
Week/Year
DRAM Manuf.
Samsung
Ranks
Dual
Part Number
CMK64GX4M4B3600C18
Correction
Serial Number
Registered
Timings Table
JEDEC #6
JEDEC #7
JEDEC #8
XMP-3596
Frequency
1037 MHz
1066 MHz
1066 MHz
1798 MHz
CAS# Latency
14.0
15.0
16.0
18.0
RAS# to CAS#
14
15
15
19
RAS# Precharge
14
15
15
19
tRAS
35
36
36
39
tRC
49
50
50
58
Command Rate
Voltage
1.20 V
1.20 V
1.20 V
1.350 V
CPU-Z
Ver. 1.90.0.x64
Tools
Validate
Close


CPU-Z
CPU
Mainboard
Memory
SPD
Graphics
Bench
About
Memory Slot Selection
Slot #2
DDR5
Module Size
32 GBytes
Max Bandwidth
DDR5-6200 (3100 MHz)
SPD Ext.
EXPO
Module Manuf.
Corsair
Week/Year
35 / 23
DRAM Manuf.
SK Hynix
Buffered
Part Number
CMP64GX5M2X6000Z30
Correction
Serial Number
Registered
Timings Table
JEDEC #6
JEDEC #7
EXPO-6000
EXPO-6200
Frequency
2400 MHz
2400 MHz
3000 MHz
3100 MHz
CAS# Latency
40.0
42.0
30.0
32.0
RAS# to CAS#
40
40
36
38
RAS# Precharge
40
40
36
38
tRAS
77
77
76
80
tRC
117
117
112
118
Command Rate
Voltage
1.10 V
1.10 V
1.400 V
1.400 V
CPU-Z
Ver. 2.08.0.x64
Tools
Validate
Close

Memorias Comerciales

TIMETEC

2 PACK



32GB
2*16G

DDR4
PC4-19200

2400MHz
CL17

NON-ECC
Unbuffered

Memorias Comerciales

TIMETEC

2 PACK



16GB
2*8G

DDR3
PC3-10600

1333MHz
CL9

ECC
Registered

Prefijos para múltiplos binarios

https://physics.nist.gov/cuu/Units/binary.html

- **Prefixes for Binary Multiples:** In December 1998 the International Electrotechnical Commission (IEC), the leading international organization for worldwide standardization in electrotechnology, approved as an IEC International Standard names and symbols for prefixes for binary multiples for use in the fields of data processing and data transmission. The prefixes are as follows:

Prefijos para múltiplos binarios

https://physics.nist.gov/cuu/Units/binary.html

Ki, Mi,

Gi, Ti,

Pi, Ei,

Zi, Yi

Factor	Name	Symbol	Origin	Derivation
2^{10}	kibi	Ki	kilobinary: $(2^{10})^1$	kilo: $(10^3)^1$
2^{20}	mebi	Mi	megabinary: $(2^{10})^2$	mega: $(10^3)^2$
2^{30}	gibi	Gi	gigabinary: $(2^{10})^3$	giga: $(10^3)^3$
2^{40}	tebi	Ti	terabinary: $(2^{10})^4$	tera: $(10^3)^4$
2^{50}	pebi	Pi	petabinary: $(2^{10})^5$	peta: $(10^3)^5$
2^{60}	exbi	Ei	exabinary: $(2^{10})^6$	exa: $(10^3)^6$

Examples and comparisons with SI prefixes	
one **kibibit**	1 Kibit = 2^{10} bit = **1024 bit**
one **kilobit**	1 kbit = 10^3 bit = **1000 bit**
one **mebibyte**	1 MiB = 2^{20} B = **1 048 576 B**
one **megabyte**	1 MB = 10^6 B = **1 000 000 B**
one **gibibyte**	1 GiB = 2^{30} B = **1 073 741 824 B**
one **gigabyte**	1 GB = 10^9 B = **1 000 000 000 B**

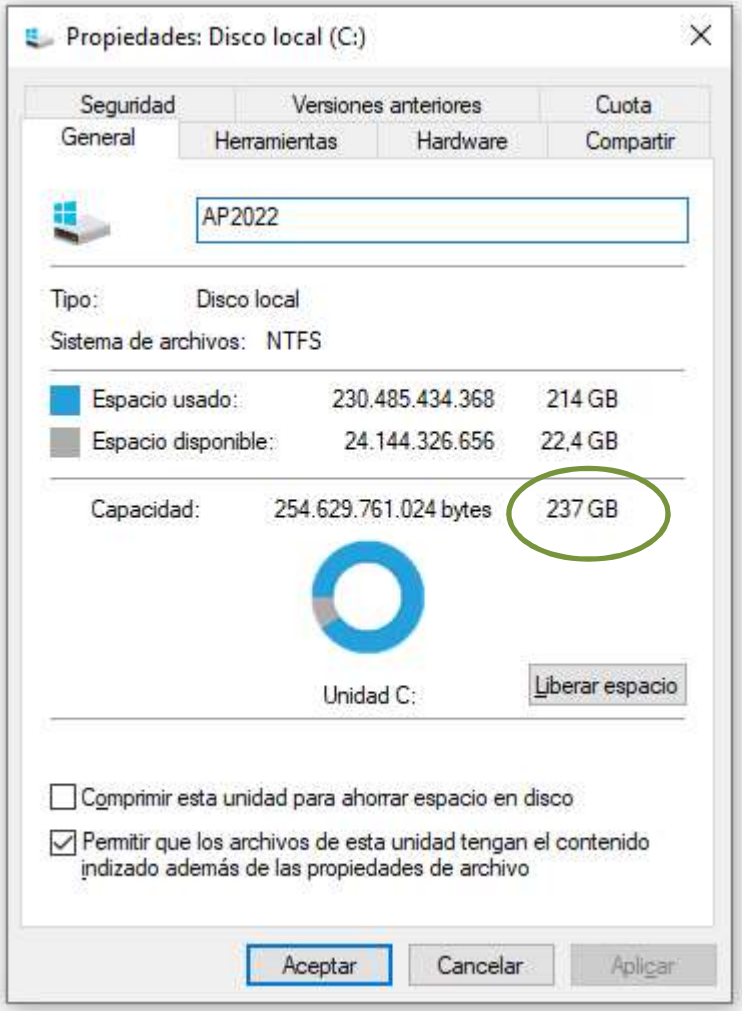
Prefijos para Múltiplos Binarios

- ROBIBYTE Y QUEBIBYTE: para la gran cantidad de datos digitales que se esperan en un futuro

Decimal term	Abbreviation	Value	Binary term	Abbreviation	Value	% Larger
kilobyte	KB	10^3	kibibyte	KiB	2^{10}	2%
megabyte	MB	10^6	mebibyte	MiB	2^{20}	5%
gigabyte	GB	10^9	gibibyte	GiB	2^{30}	7%
terabyte	TB	10^{12}	tebibyte	TiB	2^{40}	10%
petabyte	PB	10^{15}	pebibyte	PiB	2^{50}	13%
exabyte	EB	10^{18}	exbibyte	EiB	2^{60}	15%
zettabyte	ZB	10^{21}	zebibyte	ZiB	2^{70}	18%
yottabyte	YB	10^{24}	yobibyte	YiB	2^{80}	21%
ronnabyte	RB	10^{27}	robibyte	RiB	2^{90}	24%
queccabyte	QB	10^{30}	quebibyte	QiB	2^{100}	27%

Computer Organization and Design RISC-V Edition: The Hardware Software Interface

Prefijos para múltiplos binarios



En este caso
GB
debería ser
GiB
(binary value)

https://www.seagate.com/es/es/support/kb/storage-capacity-measurement-standards-002046en/

Prefijos para múltiplos binarios

https://www.seagate.com/es/es/support/kb/storage-capacity-measurement-standards-002046en/

/dev/sda - GParted

GParted Editar Ver Dispositivo Partición Ayuda

/dev/sda (19.99 GiB)

/dev/sda1 7.35 GiB | /dev/sda5 3.06 GiB | /dev/sda6 3.18 GiB | /dev/sda7 3.16 GiB | /dev/sda8 3.24 GiB

Partición	Sistema de archivos	Etiqueta	Tamaño	Usado	Libre	Opciones
/dev/sda1	ntfs		7.35 GiB	5.21 GiB	2.15 GiB	boot
/dev/sda2	extended		12.64 GiB	---		
/dev/sda5	ext4	EstanterM-CM--a 1	3.06 GiB	117.76 MiB	2.94 GiB	
/dev/sda6	ntfs	EstanterM-CM--a 2	3.18 GiB	16.82 MiB	3.16 GiB	
/dev/sda7	reiserfs	EstanterM-CM--a 3	3.16 GiB	32.19 MiB	3.13 GiB	
/dev/sda8	jfs		3.24 GiB	13.84 MiB	3.23 GiB	

0 operaciones pendientes

root@ndar: ~

```
root@ndar:~# fdisk -l /dev/sda

Disk /dev/sda: 500.1 GB, 500107862016 bytes
255 heads, 63 sectors/track, 60801 cylinders, total 976773168 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disk identifier: 0x62ff371f

   Device Boot      Start         End      Blocks   Id  System
/dev/sda1   *        2048      206847      102400    7  HPFS/NTFS/exFAT
/dev/sda2          206848   174082047    86937600    7  HPFS/NTFS/exFAT
/dev/sda3       174082110   976771071   401344481    f  W95 Ext'd (LBA)
/dev/sda5       174082112   383792849   104855369    7  HPFS/NTFS/exFAT
/dev/sda6       383793152   590256127   103231488    7  HPFS/NTFS/exFAT
/dev/sda7       590258176   844883967   127312896    7  HPFS/NTFS/exFAT
/dev/sda8       968599552   976771071     4085760   82  Linux swap / Solaris
/dev/sda9       844886016   968591359    61852672   83  Linux

Partition table entries are not in disk order
root@ndar:~#
```

Prefijos para múltiplos binarios

32 MiByte = _____ Byte

8 PiByte = _____ Byte

1 TiByte = _____ MiByte

1 MHz = _____ Hz